



Klarna Payments for SFRA

Version 26.2.0



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1. Summary

The **Klarna Payments SFRA cartridge** enables integration of Klarna Payment solution on Commerce Cloud Storefront. The integration provides merchants the flexibility to offer choice of multiple Klarna Payment products on the Commerce Cloud checkout.

This document contains the instructions for a developer to install the cartridge and integrate it on the Salesforce Commerce Cloud site. The cartridge is fully compatible with the Storefront Reference Architecture (SFRA).

Merchant teams are required also to configure the cartridge with the valid merchant credentials and site configurations in Commerce Cloud Business Manager to enable Klarna payments methods in the checkout.

The integration consists of an archive, which contains the following contents:

- Cartridge called “int_klarna_payments” and “int_klarna_payments_sfra” to be imported.
- A site-template archive containing new attributes and settings.
- This document for SFRA (Klarna Payments SFRA Integration Guide).
- The integration is based on the SFRA demo store provided by Commerce Cloud.

It is a requirement that Merchant sign a contract for integration support and production go-live with Klarna.

Klarna offers a playground (test) environment, so the integration can be tested before switching to the Klarna production environment. Based on the contract, Klarna shall provide assistance with integration and testing prior to sign-off for go-live.

2. Component Overview

2.1. Functional Overview

Key Features:

- Integrate Klarna Payments using best practices on international sites based (markets in North America, Europe, Oceania)
- Enable multiple payment products for customer in Pay Now, Pay Later and Pay Over Time categories
- Fast integration/go-live with virtual card-based integration approach for settlement
- Handle Notification: pending status updates (reject/accept) for suspected orders post review
- Site managers can customize the Klarna Payments widget styling displayed in checkout, to match the style guide of merchant website(s)
- GDPR (EU) compliant checkout flow
- Multi Shipping Address support
- Supports Klarna authorize with finalize for Bank Transfer methods (Pay Now)
- Enable Onsite Messaging placements on PDP, Cart, Header, Footer, and dedicated Info page
- Enable Klarna's Express Button on Cart page (*US) for faster checkout
- BOPIS (Buy Online, Pickup in Store) support including extra merchant data
- Support for Klarna Payment Method based promotions
- Support for adjusted price promotions with Gross Tax Policy
- Support disabling Payment method for authorization rejection
- Support Auto-Capture
- Support for custom service rate limits configuration

Klarna Payment cartridge makes use of the Klarna Payments JSON REST API and a JavaScript SDK to integrate on the storefront. Klarna Payment enables consumers to choose from the different payment method products offered by Klarna. Multiple Klarna products are available within the categories “Pay Now”, “Pay Later” and “Pay Over Time”. The cartridge integration displays payment options via a widget (iframe) added inline on the billing page, referred to as

Klarna widget or just “the widget”. The widget with information about the payment method is displayed to the customer when the individual clicks on the Klarna payment method.

Customers can authorize the payment after reviewing the payment method terms and clicking Place Order button. The Klarna order is created, and customer is re-directed to the confirmation page.

Orders successfully placed with Klarna return a Fraud Status: ACCEPTED and displayed in Business manager (BM).

With ACCEPTED status, order creation in SCC proceeds as usual. Klarna payment status is saved in a custom attribute with id kpFraudStatus in the PaymentTransaction system object, and can be seen in BM on the order details Payment tab as below:

[Merchant Tools](#) > [Ordering](#) > [Orders](#) > Order: 00005307(RefArchGlobal)

General Attributes **Payment** Notes History

Payment Information for Order '00005307'

Order Total:	£25.19
Amount Paid:	£0.00
Balance Due:	£25.19

Invoice Number:	00024502
Payment Status:	Paid

Payment Method:	KLARNA_PAYMENTS Processor: KLARNA_PAYMENTS Transaction: 40ab1ccf-1a51-266b-98a2-0d00fb59261d Amount: £25.19 Klarna Payment Category ID: pay_over_time Klarna Payment Category Name: Buy now, pay later Fraud Status: ACCEPTED
-----------------	--

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Figure 1 Klarna Payment Details in BM

An alternate flow when PENDING status is returned for Klarna order creation, the SCC order creation proceeds with modified statuses. If later a Klarna notification with updated fraud status FRAUD_RISK_ACCEPTED is returned, SCC order status is updated and returns to the usual flow. Klarna payment status is saved in a custom attribute with id kpFraudStatus in the PaymentTransaction system object, and can be seen in BM on the order details Payment tab as below:

[Merchant Tools](#) > [Ordering](#) > [Orders](#) > Order: 00005338(RefArchGlobal)

[General](#) [Attributes](#) **[Payment](#)** [Notes](#) [History](#)

Payment Information for Order '00005338'

Order Total:	£191.99
Amount Paid:	£0.00
Balance Due:	£191.99
Invoice Number:	00024524
Payment Status:	Paid
Payment Method:	KLARNA_PAYMENTS Processor: KLARNA_PAYMENTS Transaction: 08aef0f0-22f0-2138-9334-978b637d74dc Amount: £191.99 Klarna Payment Category ID: pay_over_time Klarna Payment Category Name: Buy now, pay later Fraud Status: FRAUD_RISK_ACCEPTED

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Figure 2 Klarna Payment Details in BM

2.2. Locales

The cartridge supports most locales including:

- English
- German
- Danish
- Spanish
- Finnish
- French
- Italian
- Dutch
- Polish
- Mexican

For a list of more [supported](#) locales, contact Klarna.

2.3. Use Cases

2.3.1. Enable Klarna Payments Across International Sites (NA, EU, OC)

Klarna Payments on SFRA can be configured independently on each site by locale.

2.3.2. Multitude of Payment Options for Customers

On the checkout billing step, configured Klarna payment options are dynamically loaded based on customer's cart information and the payment method categories returned for the current Klarna session.

The screen below shows an example of the payment options displayed in the Checkout billing step:

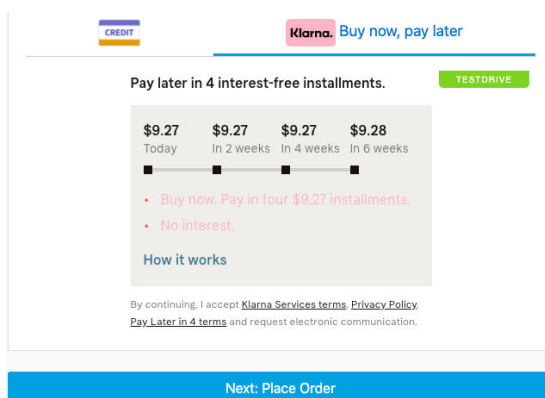


Figure 3 Payment Option

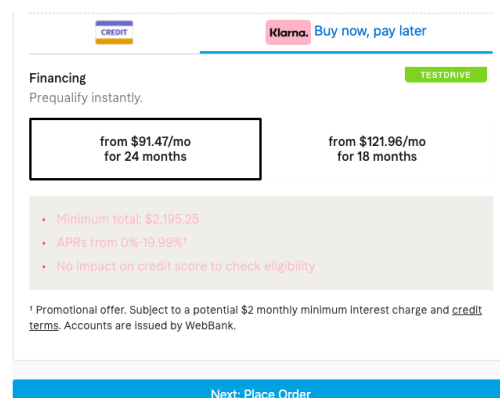


Figure 4 Payment Option

When customer selects (clicks) payment method, a widget with additional information of the Klarna product is displayed as shown above.

Note: The payment methods displayed are based on market availability and/or contractual agreement with Klarna.

2.3.3. Klarna Display Conditions and Authorization Handling

2.3.3.1. Display Conditions

Klarna payment methods may not be displayed under the following conditions:

1. **Payment Categories Not Returned:** Klarna is not displayed if payment categories are not returned in the session response.
2. **Session Linking Issue:** Payment method content is not displayed if we have payment methods in the session, but no session is linked to the basket.
3. **Authorization Denial:** Payment method category is hidden or greyed out if `show\_form` is returned as false in the Klarna response on an authorization request and the payment method is denied.
4. **Session Creation Failure:** Overall, if Klarna is unavailable and a session is not created, payment categories will not be displayed.

2.3.3.2. Authorization Handling Preference

We've implemented a site preference, `kpRejectedMethodDisplay`, allowing merchants to choose how to handle authorization rejections (refer to section 2.3.5 Refusal of Klarna Payments on Payment Method – Authorization). The available options are:

- **No action:** No specific handling for authorization rejections.
- **Hide:** Hide the payment method upon authorization rejection.
- **Gray out:** Gray out the payment method upon authorization rejection.

2.3.4. Authorize and Place Klarna Order in Checkout

Cartridge provides best practice implementation and options to include Extra Merchant Data (EMD) to optimize acceptance rate for Klarna Payment (methods) products. This includes customer info and (Buy Online, Pickup in Store) BOPIS details included as merchant data when custom site preference “attachments” is enabled for the site. The EMD data sent can be extended but should be reviewed case by case and optimized and validated based on merchant data & privacy requirements prior to go-live.

The customer's information (personally identifiable information) is sent once customer chooses a Klarna payment method and authorizes (clicks: “Continue to Place Order”). The required customer information facilitates assessment and verification of customer data to display payments method options available for the customer.

When customer clicks “Continue to Place Order” button, the authorization is initiated, and a successful authorization takes customer to summary page. Note that customer may be prompted for additional information prior to completion of authorization based on the payment method selected and market. If the authorization is not successful (“approved = false”), then the customer

will stay on the billing page, the “Continue to Place Order” button will be disabled and depending on the Business Manager settings the payment option may be hidden or grayed out (refer to section 2.3.5 Refusal of Klarna Payments on Payment Method – Authorization).

In some cases, the Pay Now payment category requires additional “finalize” call to be triggered when the customer clicks on the “Place Order” button in the review page. This is needed to ensure that the funds will be transferred (e.g.: Bank Transfer payment methods which initiate fund transfer) only when the customer has decided to place the order on the merchant site.

Please, note that in such cases cookie called “**selectedKlarnaPaymentCategory**” is set in the customer’s browser. This cookie contains the selected payment method from the previous step and is used to initialize the “finalize” call to Klarna.

For more information regarding the authorize & finalize calls, refer to the developer documentation [here](#)

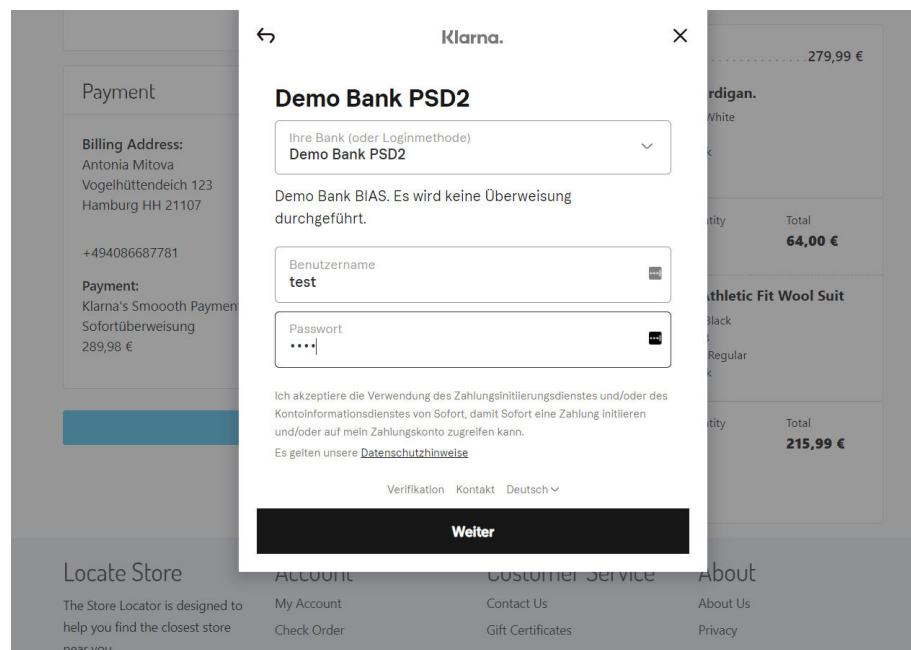


Figure 5 Finalize call screen

The Klarna Order is only placed prior to final step before SFCC order is created.

Successful payment authorizations (Klarna Payments authorization status: APPROVED) followed by placement of Klarna and SCC orders leads to a “Paid” order payment status and “Ready for Export” export status. Refused and pending Klarna Payment orders (Klarna Payments order statuses REJECTED and PENDING) lead to a “Not Paid” order payment status, and “Not Exported” export status.

In case of VCN settlement errors, Klarna orders are cancelled and SFCC orders are failed and set to a “Not Exported” export status.

When customers have selected Klarna payment option as a payment method for the order, successfully authorized the amount on order and later return to the billing page to choose a different payment method that is non-Klarna payment method. In such scenarios, when customer authorizes (non-Klarna payment method) and reaches the review page – automatic “**cancelAuthorization**” call will be triggered to release the authorized funds (Klarna related) & free up the available purchase amount for this customer.

Merchants have the option to utilize this function to cancel prior authorization when required for specific use-cases apart from the above scenario. If enabled, the checkout flow should be tested thoroughly as part of integration, considering the valid checkout scenarios (e.g. relevant Klarna session flow, Klarna payment method switching, order amount updates, checkout with external payment method, etc.).

```
36 server.get('SaveAuth', function (req, res) {  
37   ...var KlarnaSessionManager = require('*/cartridge/scripts/common/klarnaSessionManager');  
38   ...// var processor = require('*/cartridge/scripts/payments/processor');  
39   ...  
40   ...var token = req.headers['x-auth'];  
41   ...var finalizeRequired = req.headers['finalize-required'];  
42   ...  
43   ...// Cancel any previous authorizations  
44   ...// processor.cancelAuthorization();  
45   ...  
46   ...var klarnaSessionManager = new KlarnaSessionManager();  
47   ...klarnaSessionManager.saveAuthorizationToken(token, finalizeRequired);  
48   ...  
49   ...res.statusCode(200);  
50 });
```

Figure 6 Cancel Authorization Call

2.3.5. Refusal of Klarna Payments on Payment Method – Authorization

Upon selecting one of Klarna’s options as the payment method on billing step of checkout and based on the customer information provided, Klarna Payments can be refused as a payment method.

If the payment method was rejected with “**show_form=false**” & “**approved=false**” (i.e., hard reject), the merchant has the option to choose what happens with the payment option display in Billing page using BM preference “Hide Payment Methods on Deny” (**kpRejectedMethodDisplay**):

No – Leave the payment visible to the customers

Hide – The payment option will be hidden from customer

Grey Out – The payment option will be greyed out and not clickable

Please note that reloading the page will show the denied Klarna payment method again.

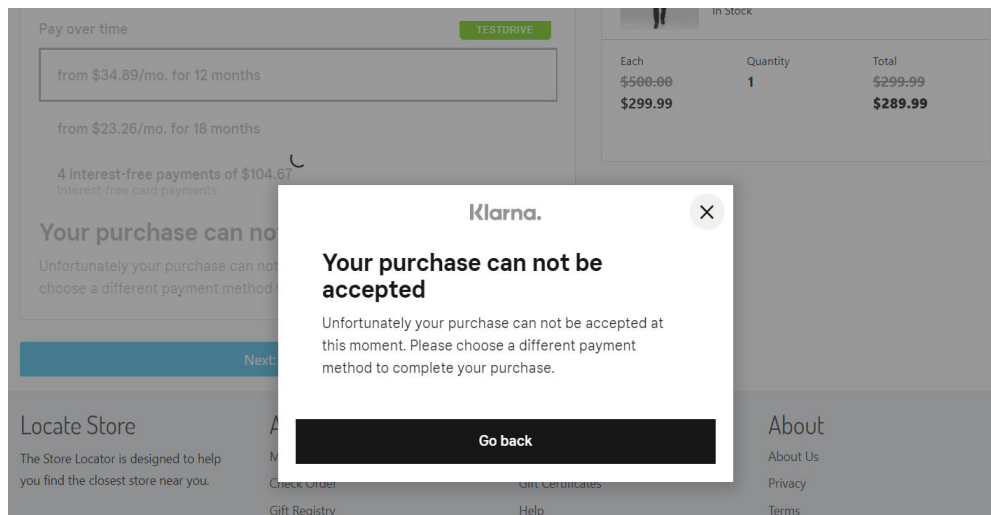


Figure 7 Denied Order Popup

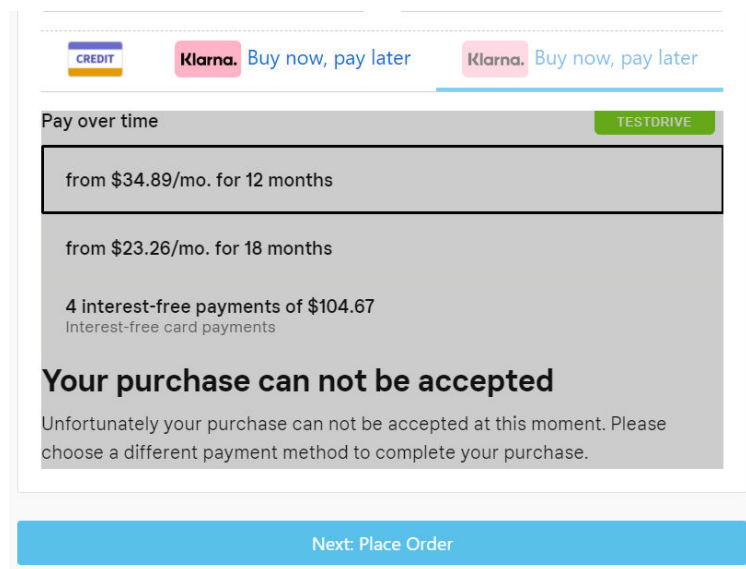


Figure 8 Greyed Out Payment Option

2.3.6. Klarna Payment Option Not Available for Current Purchase - Billing Page

The customer is presented with an appropriate message in the Klarna widget when customer attempts to choose Klarna in the billing step.

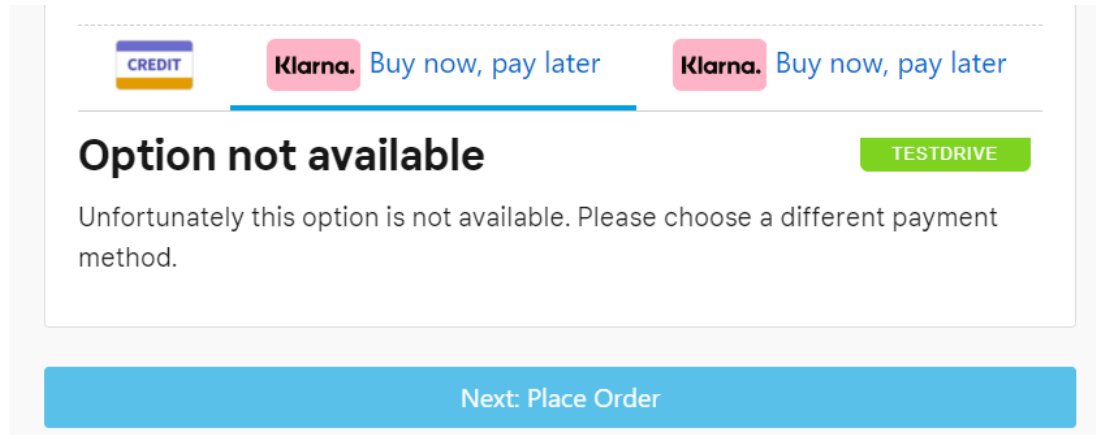


Figure 9 Option Not Available

2.3.7. Klarna Payments Not Available - Checkout

If Klarna API is not available or the site/storefront is not applicable, Klarna is not presented as a payment option in the billing page. It is recommended that Klarna session must not be created when customer chooses a non-Klarna market or merchant store, as well as in cases when merchants have multicurrency storefront with basket currency that is not supported by Klarna.

2.3.8. Handling Notifications

In scenarios where the Klarna Order has been created but instead of immediately accepting the order, the Klarna is flagged for additional review. This results in Commerce Cloud order staying in “Created” status with Fraud Status: PENDING. This order is marked with EXPORT_STATUS_NOTEXPORTED, confirmation status NOTCONFIRMED and NOTPAID.

For Klarna orders with Fraud Status PENDING, once review is complete, updates are sent to the pre-configured notification_url on the merchant Commerce Cloud site. The updated Fraud status depends on the fraud screening, the returned Fraud Status (e.g., FRAUD_RISK_ACCEPTED) is displayed in BM. The push notification is repeatedly sent (up-to 24 hours, every 10 mins) until the POST request is acknowledged with a 200 response.

Klarna sends one of the following event types in the notification to SFCC to update risk status: FRAUD_RISK_ACCEPTED, FRAUD_RISK_REJECTED, FRAUD_RISK_STOPPED.

The notification updates are generally received within 4-24 hours. The order's payment transaction is updated (see **kpFraudStatus**). This can be seen in BM on the order details Payment tab as below:

[Merchant Tools](#) > [Ordering](#) > [Orders](#) > Order: 00005338(RefArchGlobal)

General Attributes **Payment** Notes History

Payment Information for Order '00005338'

Order Total:	£191.99
Amount Paid:	£0.00
Balance Due:	£191.99
Invoice Number:	00024524
Payment Status:	Paid
Payment Method:	KLARNA_PAYMENTS Processor: KLARNA_PAYMENTS Transaction: 08aef0f0-22f0-2138-9334-978b637d74dc Amount: £191.99 Klarna Payment Category ID: pay_over_time Klarna Payment Category Name: Buy now, pay later Fraud Status: FRAUD_RISK_ACCEPTED

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Figure 10 Fraud Status in Payment Details

If the order is “FRAUD_RISK_ACCEPTED” upon notification, the order will be placed in SFCC (order status changes to OPEN). The order is marked with status CONFIRMATION_STATUS_CONFIRMED and export status EXPORT_STATUS_READY. If the order was placed with auto-capture, the payment status will be set to PAYMENT_STATUS_PAID and the full order amount will be captured.

If the order is “FRAUD_RISK_REJECTED” or “FRAUD_RISK_STOPPED” upon notification, the order is failed (FAIL) in SFCC.

Note: The Klarna pending functionality availability is dependent on markets and enabled based on the contractual agreement with Klarna.

2.3.9. Virtual Cards Settlements

This option is disabled by default. However, if standard order management is not a reasonable option for a Klarna integration, then Klarna’s Merchant Card Service based virtual card solution may be utilized via site preference “**kpVCNEnabled**”:

The screenshot displays the 'Merchant Tools' interface for a Klarna integration. It features a table of site preferences. The 'Virtual Card Network Enabled' preference is highlighted with a red border. This preference is currently set to 'Yes' and has a default value of 'No'. The table also includes other preferences like 'Secondary text color preference', 'Border Radius Preference', 'Attachments', and 'Not available message on billing page'.

Name	Value	Default Value	
Secondary text color preference	CSS hex color to be used in Klarna Payments iFrame	#999999	Edit Across Sites
Border Radius Preference	30 Size (in pixels) of the border radius to be used in Klarna Payments iFrame	5px	Edit Across Sites
Attachments	Yes Flag to switch on/off the using of attachments when creating a session	No	Edit Across Sites
Not available message on billing page	<pre>{ "GB":"Klarna Payment not available", "FR":"String in French", "AT":"Klarna Rechnung nicht verfügbar", "default":"Klarna Payment not available" }</pre> The Klarna Payments not available message on billing page. JSON string holding co...		Edit Across Sites
Virtual Card Network Enabled	Yes If set to true SFCC will create Virtual Card Network settlement from every Klarna or...	No	Edit Across Sites

Figure 11 VCN Enablement Setting

The virtual card (see note) is issued against an Klarna order, for the purpose of capturing the Klarna authorized order amount using the standard card rails.

When a customer places an order, the order is first booked in SFCC. Once an order has been accepted by Klarna, the Klarna cartridge integration creates a virtual card-based settlement, utilizing the merchant card services (MCSv3) API.

Once a settlement has been created (virtual card returned), the encrypted card details are saved in SFCC and can be later utilized by the merchant OMS platform or custom PSP integration to authorize the virtual card until the Klarna order is valid. On successful fulfillment of the order, the funds on the virtual card maybe captured. (For delays in capture, or other special use cases, please speak with the Klarna Key Account Manager in advance). While Klarna is the original payment method of the order, the order amount will be settled with the merchant using the issued virtual credit card instead of direct bank account transfer.

Please review the procedures with Klarna team for details of settlement process using virtual cards.

Refer to the below code in “**scripts/payments/processor.js**” and update accordingly to the integrated cards processor.

Note: If the Klarna order has a “**fraud_status**” of “PENDING”, action is not taken on the order until receiving Klarna’s push notification that the “**fraud_status**” has changed to “FRAUD_RISK_ACCEPTED”.

The virtual card issued is limited to 1 single successful authorization per order for a given MID

For decrypting the credit card details refer to **9.3 Decrypt VCN Card Details**.

```

118  /**
119   * Call Credit Card Authorization Hook (for VCN settlement)
120   * @param {dw.order.order} order - DW Order
121   * @returns {processorResult} - authorization result
122   */
123  function callCreditCardAuthorizationHook(order) {
124      var processorResult = null;
125      var paymentInstrument = order.getPaymentInstruments(PAYMENT_METHOD)[0];
126      var paymentProcessor = PaymentMgr;
127      var paymentMethod = paymentInstrument.paymentMethod;
128      var paymentProcessor = paymentMethod.getPaymentProcessor();
129      var transactionID = paymentInstrument.getPaymentTransaction().getTransactionID();
130      var hook = 'app.payment.processor.' + CREDIT_CARD_PROCESSOR_ID;
131      if (!HookMgr.hasHook(hook)) {
132          throw new Error('File of app.payment.processor.' + CREDIT_CARD_PROCESSOR_ID + 'hook is missing or the hook is not configured');
133      }
134      processorResult = HookMgr.callHook('app.payment.processor.' + CREDIT_CARD_PROCESSOR_ID, 'Authorize', transactionID,
135      paymentInstrument, paymentProcessor);
136      return processorResult;
137  }
138
139
140
141

```

Figure 12 Credit Card Authorization Call

To utilize virtual card integration option the merchant should:

- Enable VCN option in Site Preferences as shown above
- Enter the VCN Public Key ID. Unique UUIDv4 value, which should be different for playground testing and Production (live site)

Name	Value	Default Value
The Klarna Payments not available message on billing page. JSON string holdin...		
Virtual Card Network Enabled	Yes	No
If set to true SFCC will create Virtual Card Network settlement from every Klarn...		
VCN Public Key ID	6c5b99c5-b0de-4689-b569-1ae12ec898eb	
Unique identifier for the public key used for encryption of the card data		
VCN Private Key	MIIKQIBAAKAgEAtirQ01705IOJ5GLAGvynckEYK3V3eXZMxjh+gO5X5SAG KS1a iY6mpACVNsILYRrh4K2QV1CSn8vnn1Du56bUkNfTYuWSgEusn1ewYkP2D audwJ yIYWBvKqC5QaHTYaleKUGSwAh5ix5c+o25X6wx8B8M3Rv57mQa4sWM GexLr92 bpbUOTKl73Bjs3/6cxCO9usU2Z9FC4r2RIIGD8KjIV7TVwmbDSIC8DlelpKbZl 3w OB1iFDwO8uNIE+TJYUOKpCmw8yScR+Xe3gPkL8y6M7RQOKLHLSI8LKRE IKCK LMI3rbaNtSPN+1NSLqxrRdjrmhZhmaDFkKbBlnHoZn93jYS7Z2ZaH5F0206 Your 4096 bit RSA Private Key MIIKQIBAAKAgEAtirQ01705IOJ5GLA	

Figure 13 VCN Public Key ID

- Generate a 4096-bit RSA key pair (Refer to section Generate Key Pair and Key Id for Virtual Card Settlements). Set the custom preference “**vcnPublicKey**” with the value of the public key without the header and footer lines (begin and end public key) and the custom preference “**vcnPrivateKey**” with the value of the private key without the header and footer lines (begin and end private key)

Figure 14 VCN Public & Private Keys

- Update the VCN settlement retry setting “**kpVCNRetryEnabled**” (Deprecated with v24.4.0) “**kpVCNRetry**” . By default, this is disabled. However, if enabled and in cases when a VCN creation error is returned, the application will retry the settlement request once again with order_id as the idempotency key.
- Finally, you need to send the generated unique key_id + public key combination in JWK format to Klarna prior to testing and go-live. It will be used to encrypt the aes key which is used for encrypting the pci data on Klarna side when settlement request is made. After confirmation from Klarna that the key has been successfully added to your merchant profile you would be able to use virtual card-based settlement option for Klarna payment methods

If enabled and fully configured, virtual card settlement request is made successfully. For orders placed with the VCN settlement option, the related custom attributes are shown below:

[Merchant Tools](#) > [Ordering](#) > [Orders](#) > Order: 00049309(RefArch)

General **Attributes** Payment Notes History

Attributes for Order '00049309'

On this page you can edit the attributes of the order. Fields with a red asterisk (*) are mandatory. Click **Apply** to save changes. Click **Reset** to revert your changes.

Klarna Payments

Klarna Payments Order ID:

72bf2c96-6523-2add-8c50-f2af87712019

Is VCN Used:

☒

VCN Card ID:

befbb0e9-5e98-4e39-9c00-75aba0c3372b

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Figure 15 VCN Details in Order

If required, the additional virtual card details can be assigned to this group in Administration > Site Development > System Object Types > select “Order”. In the Attribute Grouping tab select Klarna_Payments and click on “edit”. Assign the new attributes and save the data.

[Administration](#) > [Site Development](#) > [System Object Types](#) > [Order - Attribute Groups](#) > Klarna Payments

Object Type 'Order' - Attribute Definition Assignments

On this page you can assign existing attribute definitions to your attribute group.

Assign Attribute Definition

ID:*

...

Add

Select All	ID	Name	Type	Attribute Settings	Sorting
☐	kpOrderID	Klarna Payments Order ID	String		
☐	kpIsVCN	Is VCN Used	Boolean		
☐	kpVCNCardID	VCN Card ID	String		
☐	kpVCNHolder	VCN Holder	String		
☐	kpVCNBrand	VCN Brand	String		
☐	kpVCNPCIData	VCN PCI Data	String		
☐	kpVCNIV	VCN Initialization Vector	String		
☐	kpVCNAESKey	VCN AES Key	Text		

Unassign

<< Back

Figure 16 Full List of VCN Attributes

Please work with Klarna Account Manager and Delivery contact in advance to select the appropriate virtual card product based on your business requirements and use-cases. You can find information [here](#) around other use cases supported.

Important Note!

DO NOT SAVE DECRYPTED PCI DATA ON THE SERVER. It is the responsibility of the merchant to ensure PCI-DSS compliance and to ensure the card data is handled securely in co-ordination with required partners/Payment Service Provider/Acquirer. Please review in advance the order export details required for virtual card-based Klarna orders. Any historical

decrypted PCI data should also be expunged, regardless of the validity date (see section 3.5.1 Job “OrderCleanUp”).

2.3.10. Auto-Capture

Auto-capture is enabled via a site preference “**kpAutoCapture**” located in “**Klarna_Payments**” preference group.

When the preference is enabled (disabled by default), a full amount capture is attempted. If the capture is successful, the SFCC order’s payment transaction is marked as Paid, and viewable in the Business Manager.

The order will be marked as “*Captured*” in the Klarna’s Merchant Portal

Merchant reference 1

00029205

Edit

Klarna reference

BJJGH9P1

Created

Jun 25, 2019, 5:00 PM

Expires

Jul 23, 2019, 3:00 AM

Merchant ID

K500726

Customer

Shipping address

Angela Gill

4939 Wyatt Street

West Palm Beach

5942 4RG Florida

GB

Tel

01222 555 555

Email

ivan.zanev@tryzens.com

Edit shipping address

Billing address

Additional Info

Order lines (2)

Refund

Qty	Item	Reference	Unit price	Discount	Tax	Amount
1	Black Flat Front Wool Suit	7505187030...	191.99	0.00	5% 9.14	191.99
1	Наземен транспорт	GBP001	7.99	0.00	5% 0.38	7.99

PAYMENT DETAILS

Initial Payment Method

Statement

Resend statement

ORDER TOTAL

Captured

Refunded

Not Captured

CUSTOMER BILLED

E199.98

E199.98

E0.00

E0.00

E199.98

Activity Log

Jun 25, 2019

9 minutes ago

5:00 PM

Order acknowledged

Via API

5:00 PM

Captured: E199.98

Via API

5:00 PM

Order placed: E199.98

Figure 17 Order Details in Klarna Portal

If the capture is unsuccessful, an error will be logged in the custom error log. The setting must be reviewed with Klarna delivery team before testing and go-live.

Note: Auto-capture is possible for orders when VCN is not enabled!

2.3.11. Widget Customizations

Note: The merchant will need a configured Klarna Payments account.

The merchant can style the Klarna Payments widget (skin), to match the marketing and branding needs of their store. The list with the graphic elements that can be customized out of the box through site preferences are listed below:

"color_details" (site preference kpColorDetails): "#C0FFEE"
"color_button" (site preference kpColorButton): "#C0FFEE"
"color_button_text" (site preference kpColorButtonText): "#C0FFEE"
"color_checkbox" (site preference kpColorCheckbox): "#C0FFEE"
"color_checkbox_checkmark" (site preference kpCheckboxCheckmark): "#C0FFEE"
"color_header"(site preference kpColorHeader): "#C0FFEE"
"color_link"(site preference kpColorLink): "#C0FFEE"
"color_border"(site preference kpColorBorder): "#C0FFEE"
"color_border_selected"(site preference kpBorderSelected): "#C0FFEE"
"color_text"(site preference kpColorText): "#C0FFEE"
"color_text_secondary"(site preference kpColorTextSecondary): "#C0FFEE"
"radius_border"(site preference kpRadiusBorder): "0px"

Note : Site preferences kpColorDetails, kpColorButton, kpColorButtonText, kpColorCheckbox, kpCheckboxCheckmark, kpColorHeader, kpColorLink, kpColorBorder, kpBorderSelected, kpColorText, kpColorTextSecondary, kpRadiusBorder are deprecated .A new site preference kpColorCustomization is created under klarna_KP which includes all the parameters mentioned in a JSON

2.3.12. Customizing Payment Method Name

The payment method name “*Klarna Payments*” may be customized via the “*Merchant Tools > Ordering > Payment Methods*” section in Business Manager.

The screenshot below shows the “*Klarna*” method selected and the administrator choosing a language from the drop-down.

Payment Methods

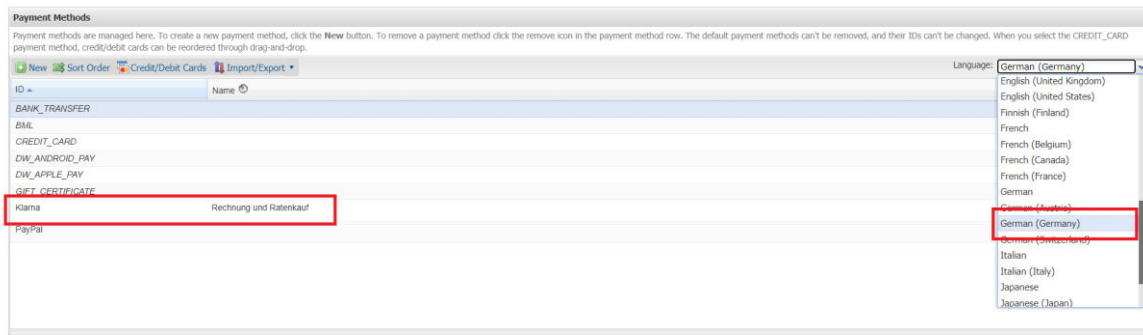


Figure 18 Customize Payment Name

The payment method name would then be visible in the mini summary & confirmation screens, the confirmation emails and My Account Order Details section.

Order Number: 00055604

Payment Information		
Billing Address	Payment Method	Payment Total
Testperson-de	Rechnung und Ratenkauf	Subtotal 112
Approved	Klarna order reference: 70aa2465-33dd-2ee0-b3f9-	Order Discount - 11
Vogelhüttendeich 123	5a038dbe0a71	€
Hamburg, HH 21107	Amount: 108,79 €	Shipping 7,9€
Germany		Ground
Phone: +494086687781		Order Total: 108

Shipment#1

Item	Quantity	Price	Shipping To
Straight Fit Shorts	1	112,00 €	Testperson-de Approved

Figure 19 Payment Method Name in Email

2.3.13. Klarna On-Site Messaging

On-site messaging is a platform that enables you to add tailored messaging to your website. With On-site messaging you can let shoppers know about the different payment options you have available as they browse your site. By using Klarna, customers have access to our flexible

payment options in the checkout; On-site messaging is a great way to let them know even before they decide to buy.

The Klarna Payment cartridge provides multiple options in the standard implementation based on the reference architecture:

- Product Page and Cart based promotions
- Sitewide
 - Top Banner strip
 - Footer logo
- Custom Info-page

Klarna On-Site Messaging (OSM) is configured by site and by locale via the **KlarnaCountries** custom object.

To configure the OSM settings for a locale, you must visit “**Merchant Tools – Custom Object Editor**” and search for **KlarnaCountries** custom object. Select the country Key, e.g. “US” .

In the custom country specific configuration, provide a valid locale for the OSM tag based on the country being configured.

The OSM Data Client ID and Data keys required are available in Klarna Merchant Portal within the On-site Messaging App:

On-site Messaging

URL to On-Site Messaging Library URL:

On-site messaging Data Client ID:

On-Site Messaging Environment:

Cart Placement Tag Enabled: ☒

Cart Placement Data Key:

Cart Placement Theme:

Cart Placement Custom Styling (#osm-cart-placement):

PDP Placement Tag Enabled: ☒

PDP Placement Data Key:

PDP Placement Theme:

PDP Placement Custom Styling (#osm-pdp-placement):

```
}  
  
#osm-pdp-placement::part(osm-heading) {  
  font-family: 'Lobster', cursive;  
}  
  
#osm-pdp-placement::part(osm-message) {  
  color: #46C2CB;  
}  
  
</style>
```

Figure 20 OSM Settings in BM

The screenshot displays the 'OSM Settings in BM (cont.)' configuration page. It is organized into three main sections, each with a 'Tag Enabled' checkbox, a 'Data Key' input field, a 'Theme' dropdown, and a 'Custom Styling' text area.

- Header Placement:**
 - Header Placement Tag Enabled: ☒
 - Header Placement Data Key:
 - Header Placement Theme:
 - Header Placement Custom Styling (#osm-header-placement):
- Footer Placement:**
 - Footer Placement Tag Enabled: ☒
 - Footer Placement Data Key:
 - Footer Placement Theme:
 - Footer Placement Custom Styling (#osm-footer-placement):

```
<style>
#osm-footer-placement::part(container) {
  background-color: #FAEAB1;
}

#osm-footer-placement::part(badge) {
  transform: scale(1.1);
}
</style>
```
- Info Page Placement:**
 - Info Page Placement Tag Enabled: ☒
 - Info Page Placement Data Key:
 - Info Page Placement Theme:
 - Info Page Placement Custom Styling (#osm-info-page-placement):

```
<style>
@import url('https://fonts.googleapis.com/css2?family=Lobster&family=Montserrat:ital,wght@1,100&display=swap');

#osm-info-page-placement::part(osm-container){
  font-family: 'Montserrat', sans-serif;
  color: #F49D1A;
}

#osm-info-page-placement::part(osm-heading) {
  font-family: 'Lobster', cursive;
}
```

At the bottom, there is a checkbox for 'Data Inline Enabled on PDP/Cart placement (Canada only):' which is also checked.

Figure 21 OSM Settings in BM (cont.)

To enable Placement tag for the Cart Page, the “Cart Placement Data Key” must be filled with Data Key value and the “Cart Placement Tag Enabled” must be checked. To customize the Cart Placement – theme could be selected and/or custom CSS entered in “Cart Placement Custom Styling” attribute using the placement id - **#osm-cart-placement**.

Note: Cart placements amount must be [updated](#) and latest Klarna credit offering placement (where required) displayed to customer when order line quantity is updated on cart page.
Refer to **3.6.2** app_storefront_base\cartridge\client\default\js\cart\cart.js (required)

To enable Placement tag for the PDP Page, the “PDP Placement Data Key” must be filled with Data Key value and the “PDP Placement Tag Enabled” must be checked. To customize the PDP Placement – theme could be selected and/or custom CSS entered in “PDP Placement Custom Styling” attribute using the placement id - **#osm-pdp-placement**.

To enable Placement tag for header, the “Header Placement Data Key” must be filled with Data Key value and the “Header Placement Tag Enabled” must be checked. To customize the Header Placement – theme could be selected and/or custom CSS entered in “Header Placement Custom Styling” attribute using the placement id - **#osm-header-placement**.

To enable Placement tag for footer, the “Footer Placement Data Key” must be filled with Data Key value and the “Footer Placement Tag Enabled” must be checked. To customize the Footer

Placement Tag – theme could be selected and/or custom CSS entered in “Footer Placement Custom Styling” attribute using the placement id - **#osm-footer-placement**.

To enable Placement tag for the Info Page, the “Info Page Placement Data Key” must be filled with Data Key value and the “Info Page Placement Tag Enabled” must be checked. To customize the Info Page Placement Tag – theme could be selected and/or custom CSS entered in “Info Page Placement Custom Styling” attribute using the placement id - **#osm-info-page-placement**.

In Library URL, please input the full URL to the On-Site Messaging JavaScript Library.

In On-Site Messaging Environment please select the corresponding environment playground for test or production.

In On-site messaging Data Client ID, please input the On-Site Messaging Data client ID value.

On-site messaging Data locale, please input the valid On-Site Messaging Data locale.

All the placements can be customized with custom CSS using the placement component id – in the brackets of attribute name. The customization value should be wrapped in `<style>` element:

Footer Placement Custom Styling (#osm-footer-placement):

```
<style>
#osm-footer-placement::part(container) {
  background-color: #FAEAB1;
}

#osm-footer-placement::part(badge) {
  transform: scale(1.1);
}
</style>
```

Customization css

Figure 22 OSM Placements CSS customization

Note: Current configuration of using attributes from Custom Object is deprecated with version 24.3. In the new implementation, settings are retrieved from Site Preferences as explained below

Name	Value	Default Value
osm_enable Add personalized messaging throughout the shopper journey for higher conversion rates and increased spend.	Yes	Yes
Theme* (osm_theme)	Light (recommended) (Light (recommended))	
Placement* (osm_placement)	None Product Page (productPage) Cart Page (cartPage) Site Wide Banners (siteBanners) FAQ (faq) Footer (footer)	
Placement (osm_custom_styling)		

Fig 23 OSM configuration in BM

Current configuration under BM is placed in **Merchant Tools – Site Preferences-Custom Preference under Klarna_OSM group**

To enable Klarna OSM implementation in storefront ,**osm_enable** must be set to true.

osm_theme dropdown is used to apply theme to OSM .Options include default,dark and custom.

To enable placement in pdp,card,header,footerand info page **osm_placement** is enabled in the mutiselect dropdown. Placement tag ID is hardcoded from code .

Placement	TagId
Product Page	credit-promotion-auto-size
Cart Page	credit-promotion-badge
Site Wide Banner	top-strip-promotion-badge
FAQ	info-page-auto-size
Footer	footer-promotion-auto-size

Table: Placement tag id for different pages

Placement styling is taken from **osm_placement_styling** which is a JSON.

osmLibraryURL is hardcoded and retrieved from constants.

osmEnvironment is used from KlarnaPayments.

CSS customizations are available only for the new OSM library version.

Please follow Klarna guidance for placement styling - <https://docs.klarna.com/on-site-messaging/in-depth-knowledge/styling-on-site-messaging-with-css/>.

For Canada only, please update “Data Inline Enabled on PDP/Cart placement (Canada only)” value as follows:

- For PayBright enabled payment methods – set to true.
- For Klarna enabled payment methods – set to false.

Note : As per new implementation, osmDataInlineEnabled is deprecated

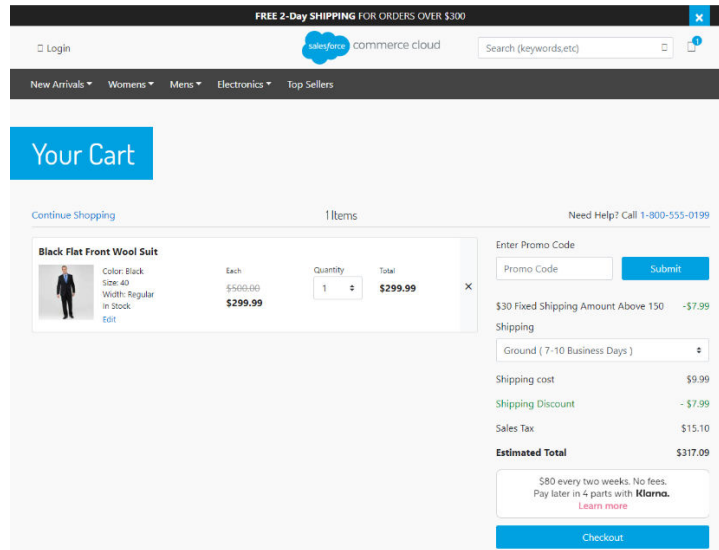


Figure 23 On-Site Messaging Enabled on Cart Page

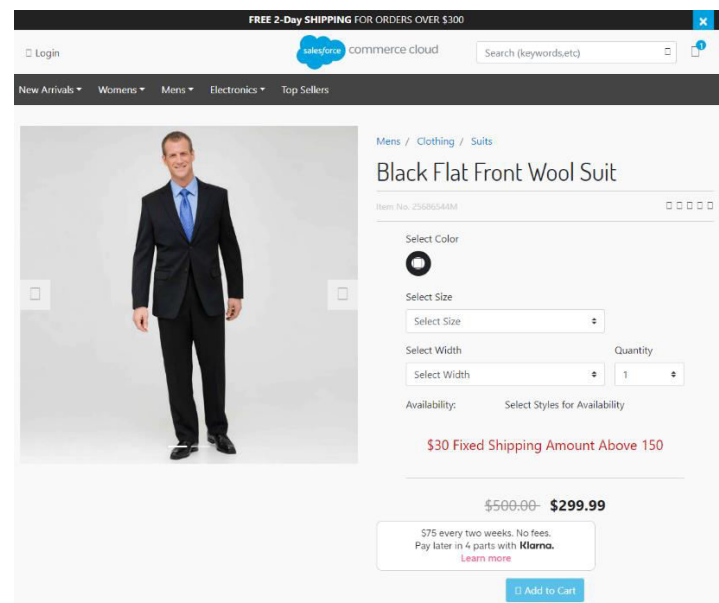


Figure 24 On-Site Messaging Enabled on PDP Page

In addition to the above, if you wish to display the dedicated (custom) Klarna info OSM page you can use the following controller endpoint “**KlarnaPayments-InfoPage**”. For example, you should update the “**footer-about**” content asset to include this line of code as shown on the screenshot.

Klarna Info

```
Body: <h3><a class="title" href="#">About</a></h3>
      <ul class="menu-footer content">
        <li><a href="$url('Page-Show', 'cid', 'about-us')$" title="Go to About Us">About Us</a></li>
        <li><a href="$url('Page-Show', 'cid', 'privacy-policy')$" title="Go to Privacy">Privacy</a></li>
        <li><a href="$url('Page-Show', 'cid', 'terms')$" title="Go to Terms">Terms</a></li>
        <li><a href="$url('Page-Show', 'cid', 'jobs-landing')$" title="Go to Jobs">Jobs</a></li>
        <li><a href="$url('KlarnaPayments-InfoPage')$" title="Go to Klarna Info">Klarna Info</a></li>
      </ul><!-- END: footer_about -->
```

Figure 25 Footer Asset Update

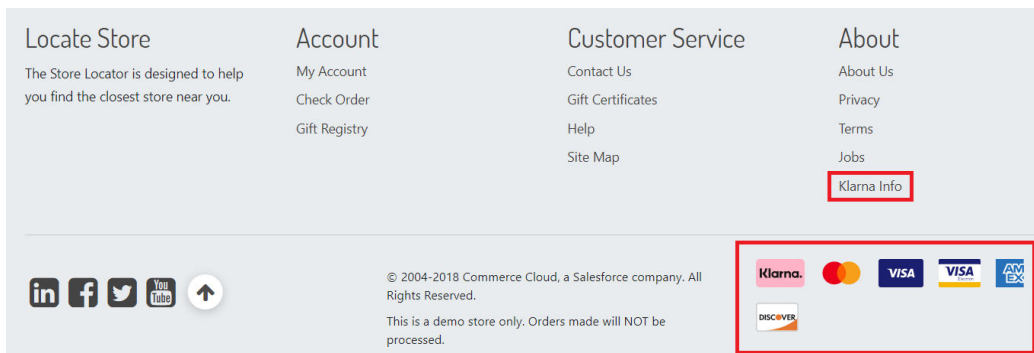


Figure 26 On-Site Messaging Enabled on Footer and link to Klarna Info Page

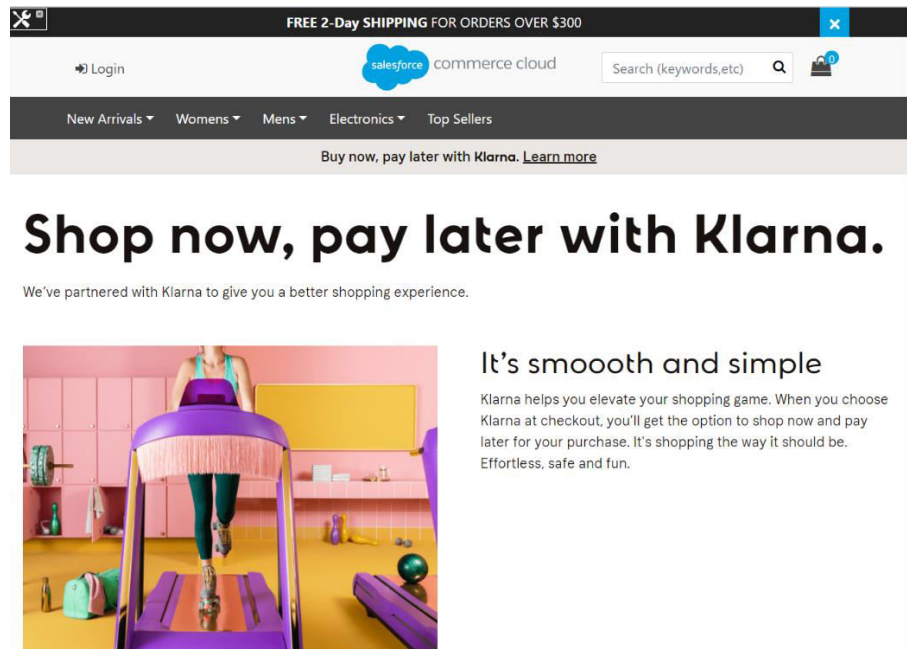


Figure 27 On-Site Messaging Enabled on Header & dedicated Klarna Info Page

For more information regarding OSM customizations and best practices, please refer to the Klarna [Developer/ Docs](#)

Integration Best practice + information about Klarna Branding and Co-marketing options [here](#).

Note: It is the merchant's responsibility to ensure that user content is collected when required for OSM placements for your local market to abide by the legal requirement. e.g., [EU cookie-guide](#) in Klarna Merchant Portal

2.3.14. Klarna Express Button

The cartridge supports the **Klarna Express button (KEB)** on the standard Cart page and mini-Cart.

By enabling Klarna's Express button on the cart page of your website, shoppers can choose to log into their (*or sign up for a*) Klarna account and have their personal details pre-filled in the checkout. Klarna's payment method will be pre-selected for the shopper. In [supported](#) markets, Klarna's network of shoppers benefit from a pre-filled checkout experience, which also includes first time shoppers on a merchant storefront.

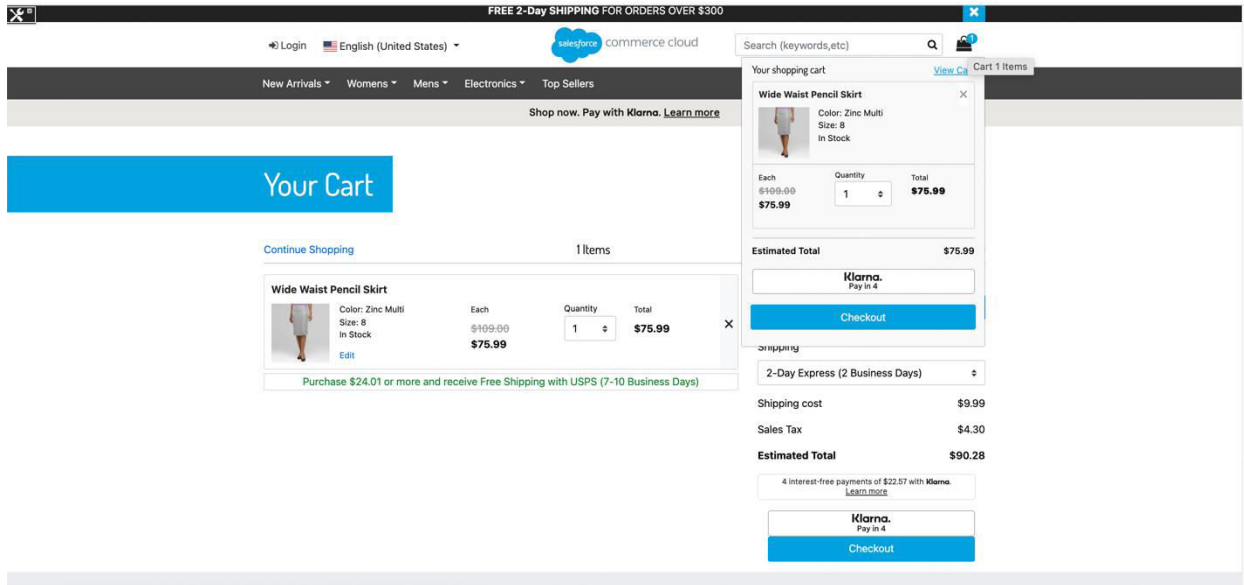


Figure 28 Klarna Express Button on Cart Page

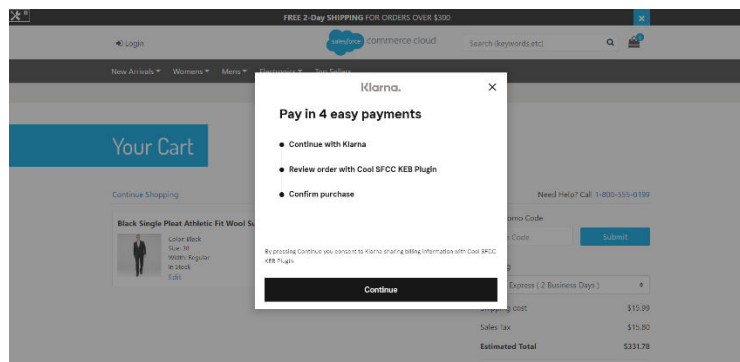


Figure 29 Klarna Express Button clicked Cart Page

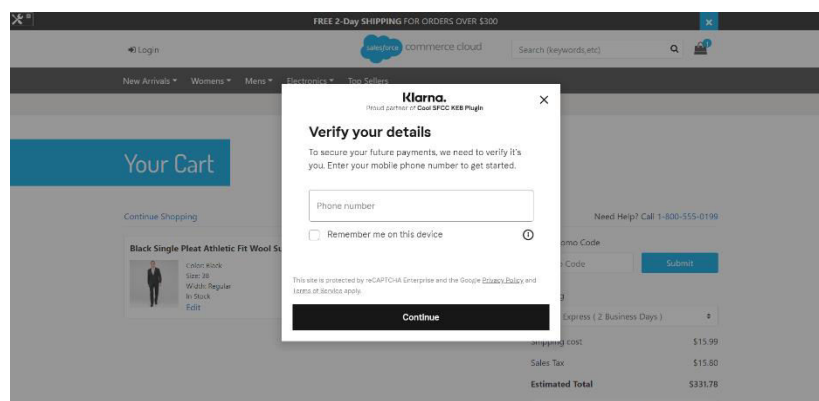


Figure 30 Klarna Express Button user credentials input actions

Once the credentials have been provided and the shopper has successfully authenticated with their Klarna account, they are redirected to the checkout page with relevant details pre-populated and Klarna Payment (e.g.: “Pay in 4”) method pre-selected on the billing page.

The screenshot displays a checkout interface with the following sections:

- Shipping:**
 - Shipping Address: John Doe, Lombard St 10, Beverly Hills, CA 90210, +16173279035
 - Shipping Method: 2-Day Express (2 Business Days) - \$15.99
- Payment:**
 - Billing Address: John Doe Lombard St 10 Beverly Hills, CA 90210
 - *Email: someone@example.com
 - *Phone Number: +16173279035
 - Selected Method: Klarna, 4 interest-free payments
 - User Profile: John Doe, someone@example.com
 - Payment Plan: 4 interest-free payments of \$80.32

Payment	Amount	Due Date
1	\$80.32	Today
2	\$80.32	In 2 weeks
3	\$80.32	In 4 weeks
4	\$80.32	In 6 weeks
- Order Summary:**
 - Subtotal: \$289.99
 - Shipping: \$15.99
 - Sales Tax: \$15.30
 - Total: \$321.28
- Items:**
 - 1 Items: \$289.99
 - Black Single Pleat Athletic Fit Wool Suit

Each	Quantity	Total
\$289.99	1	\$289.99

At the bottom, there is a blue button labeled "Next: Place Order".

Figure 31 Checkout details prefilled and Klarna Pay in 4 selected

For storefronts with custom checkout design or address field validations, please ensure that the authenticated users address data is populated without modification to ensure the Klarna Payment method authorization is completed successfully.

Note: On re-direct, in the standard checkout, any existing email ID and phone number are updated with the latest provided by the shopper

2.3.15. Klarna Payment Method Based Promotions

As of B2C 20.7 release, merchants can include payment methods as qualifier for product, order & shipping promotions.

OOTB when such promotion is set up to use a payment method as qualifier, the total order amount will be visible to the customer once they reach the review page. This means that the Klarna authorization call will be made for larger amount than the final one.

To address this issue, once the customer clicks on any of the payment options in the billing section – a call will be made to the backend. This will re-calculate the basket totals if any promotions are applicable and will update the Klarna session details. As a result, the Klarna iframe widgets and the mini summary section on the storefront will update to show the final order details.

When the selected payment method is non-Klarna one, this logic should be customized by the merchant to handle any 3rd party payment integrations.

2.3.16. Price Adjustment Taxation Handling

OOTB the Klarna API calls will send the product / shipping method details and the relevant discounts as separate lines items as shown below:

```
"order_lines": [
  {
    "type": "discount",
    "name": "5 Off Ties Promotion",
    "reference": "682875540326M_$5_off_ties_promotion",
    "quantity": 1,
    "merchant_data": "5ties",
    "unit_price": -500,
    "tax_rate": 500,
    "total_amount": -500,
    "total_tax_amount": 0,
    "total_discount_amount": 0,
    "product_url": null,
    "image_url": null
  },
  {
    "type": "physical",
    "name": "Checked Silk Tie",
    "reference": "682875540326M",
    "quantity": 1,
    "merchant_data": "",
    "unit_price": 1919,
    "tax_rate": 500,
    "total_amount": 1919,
    "total_tax_amount": 68,
    "total_discount_amount": 0,
  }
]
```

Figure 32 Line Items with Default Taxation Setting

In some cases, merchants using gross taxation might enable the “Tax Products and Shipping Only Based on Adjusted Price” preference under “Merchant Tools > Site Preferences > Pricing and Promotion” where the price adjustments are not taxed.

The setting, called “**kpPromoTaxation**” has been introduced, where merchants should update this to match the promotion setting below:

- price (Based on Price) – The product, shipping and their discounts will be sent as separate lines. This is the default setting.

- adjustment (Based on Adjusted Price) – When this is selected, the product or shipping method line item will be sent with attribute “total_amount” matching the prorated price and attribute “total_discount_amount” – matching the total sum of all discounts for this item.

```

"order_lines": [
  {
    "type": "physical",
    "name": "Checked Silk Tie",
    "reference": "682875540326M",
    "quantity": 1,
    "merchant_data": "",
    "unit_price": 1919,
    "tax_rate": 2200,
    "total_amount": 1419,
    "total_tax_amount": 256,
    "total_discount_amount": 500,
  }
]

```

Figure 33 Line Items with "Based on Adj." Taxation Setting

Note: Enabling this setting is not required for storefronts with net taxation as the tax is not included in the products base price. The total order sales tax is sent as a separate line item to Klarna and not on product/shipping line-item level.

2.3.17. Buy Online, Pickup in Store (BOPIS)

When store pickup has been enabled on the storefront, the integration will send stores details to Klarna in the authorization request and when placing the Klarna order. Store information is not sent prior the interaction of the customer with the Klarna payment method widgets.

The store address(es) is always included in the EMD attachment “other_delivery_address” when applicable.

The address included on the shipping address in the Klarna order with store pick-up, is as below:

- Orders that have 1 or more store pickup shipments (no home delivery address), the first store shipment details will be set as the shipping address
- Order with store pick-up(s) and home delivery shipment, home delivery address will be used as the shipping address in the Klarna calls
- If the order contains no store pickups, no information is sent in “other_delivery_addresses” attribute

{

```

"attachment": {
  "content_type": "application/vnd.klarna.internal.emd-v2+json",
  "body": {
    "other_delivery_address": [{
      "shipping_method": "store pick-up",
      "shipping_type": "normal",
      "first_name": "Test",
      "last_name": "Customer",
      "street_address": "1487 Bay St",
      "street_number": "",
      "postal_code": "01109",
      "city": "Springfield",
      "country": "US"
    }]
  }
}

```

For more information on the options refer [here](#)

2.3.18. Configuration Support for Service Rate Limits

Klarna Payment API sets rate limits by operation (session creation, order creation, etc) to maintain a high quality of service for all its customers. A merchant has the flexibility to request higher rate limits for a specific duration. The duration of such events may last the period of high traffic events (e.g.: Flash sales or Holiday shopping). To enable these agreed rate limits, the service rate limits feature can be utilized.

The Service ID (e.g.: klarna.http.createSession) and Service Profile (e.g.: klarna.http.createSession) for default operations are included in the meta file.

salesforce

Sandbox - zzjh

RefArch

Merchant Tools

Administration

Storefront

Toolkit

Administration > Operations > Services

Services

Profiles

Credentials

Services

Select All	Name	Type	Profile
☐	int_braintree.http.xml.payment.Braintree	HTTP	Braintree Default Profile
☐	int_paypal.http.nvp.payment.PayPal.RefArch	HTTP	PayPal Default Profile
☐	klarna.http.cancelAuthorization	HTTP	klarna.http.cancelAuthorization
☐	klarna.http.cancelOrder	HTTP	klarna.http.cancelOrder
☐	klarna.http.createCapture	HTTP	klarna.http.createCapture
☐	klarna.http.createOrder	HTTP	klarna.http.createOrder
☐	klarna.http.createSession	HTTP	klarna.http.createSession
☐	klarna.http.defaultendpoint	HTTP	klarna.http.service
☐	klarna.http.getOrder	HTTP	klarna.http.getOrder
☐	klarna.http.getSession	HTTP	klarna.http.getSession
☐	klarna.http.updateSession	HTTP	klarna.http.updateSession
☐	klarna.http.vcnSettlement	HTTP	klarna.http.vcnSettlement

Default Klarna Service Operations:

- Create Session
- Update Session
- Create Order
- Cancel Order
- Capture Order (Full Capture)
- New Order Settlement (Virtual Card)

For more information on the options refer [here](#)

2.3.19. Klarna Subscriptions

The cartridge supports subscription handling.

2.3.19.1. Configuration

Subscription details are configured on product level. Products can be subscription, standard or both. The trial period should be an integer value.

Klarna Subscription	
Is Klarna Subscription Product:	Yes
Klarna Trial Days Usage:	15 (Integer)
Is Klarna Standard Product:	No

Figure 34 Product subscription configuration

2.3.19.2. Cart page

- Subscription products

Subscription-only products are automatically added to the shopping cart as subscription line items. For products that can be both subscription and standard, users can select their preference on the cart page. of standard and subscription products with different trial periods, or if some products

have a trial period while others do not. It's important to note that the checkout process won't proceed if the cart contains a mix

Continue Shopping

3 Items

3/4 Sleeve Boat Neck Striped Knit



Color: Navy Multi

Size: L

In Stock

[Edit](#)

Each

~~\$79.00~~

\$71.10

10% off Summer Flats

Quantity

1

Total

~~\$79.00~~

\$71.10

☒ Subscribe

This product can be purchased as a Subscription with Klarna payments.

standard and subscription product without trial period

3/4 Sleeve V-Neck Top



Color: Butter

Size: L

In Stock

[Edit](#)

Each

~~\$24.00~~

\$21.60

10% off Summer Flats

Quantity

1

Total

~~\$24.00~~

\$21.60

☐ Subscribe

This product can be purchased as a Subscription with Klarna payments.

standard and subscription product with trial period

Subscription Trial period: 12 days

3/4 Sleeve Solid Drape Neck Sweater



Color: Carnation

Size: M

In Stock

[Edit](#)

Each

~~\$46.99~~

\$42.29

10% off Summer Flats

Quantity

1

Total

~~\$46.99~~

\$42.29

☒ Subscribe

This product can be purchased as a Subscription with Klarna payments.

subscription only product with trial period

Subscription Trial period: 12 days

Figure 35 Subscription products on cart page

- Subscription details

Dropdown menus with predefined values appear on the cart page when there's at least one subscription product in the cart. The configuration of these values can be managed in the Administration panel under "Site Development" > "System Object Types" > "Basket - Attribute Definitions." The attributes available for configuration are:

- kpSubscriptionFrequency
- kpSubscriptionPeriod

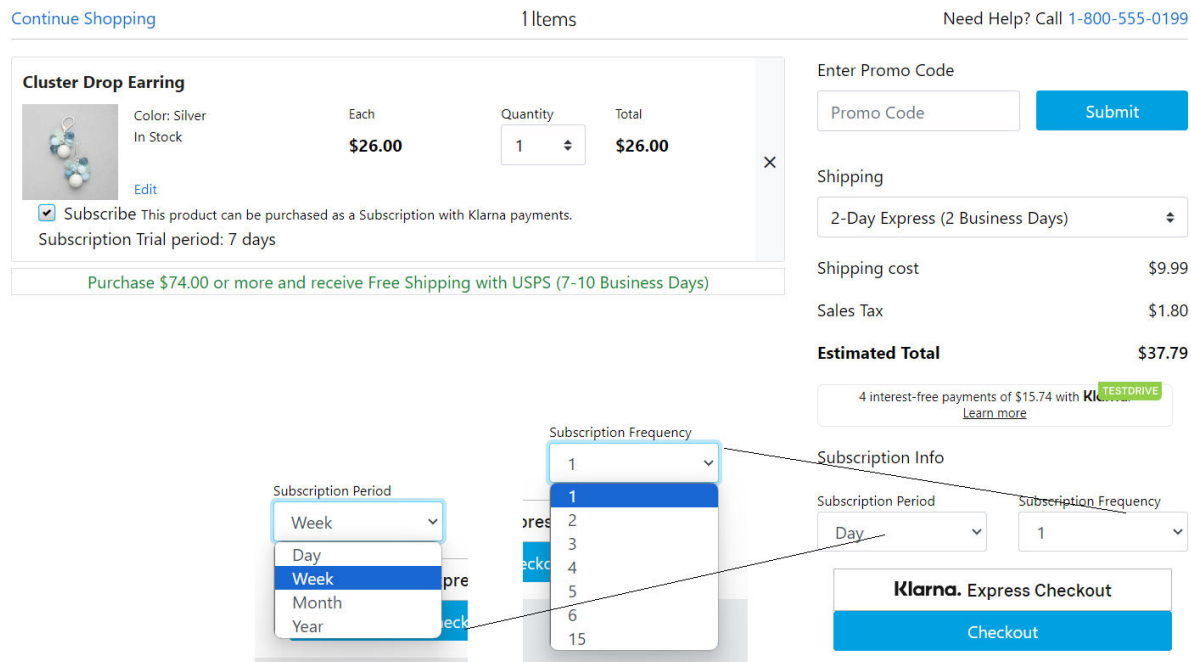


Figure 36 Subscriptions details in cart page

2.3.19.3. Checkout

Only Logged in user is allowed to do a subscription check out. Session intent is defined based on the basket content:

- “tokenize” – basket contains products with trial period, user is not charged on order creation
- “buy_and_tokenize” – subscription products without trial period
- “buy” – only standard products, no subscription products

For intents “tokenize” and “buy_and_tokenize” Klarna customer token for recurring payments is created. This is stored in customer profile for future usage.

Once the order is created the user profile is updated with subscription data:

- token
- enabled: Status of subscription
- nextChargeDate: A calculated date for the next subscription charge

- subscriptionPeriod: An enumerated value representing the subscription period ('week', 'month', or 'year').
- subscriptionFrequency: The frequency of subscription with will be numeric 1,2,3,4,5,6,15)
- subscriptionProductID: The corresponding subscription product ID.
- lastOrderID: The ID of the last order for the subscription.

2.3.19.4. Account Subscription dashboard

There is a full list of subscriptions per user in my account section. The user will be able to cancel subscriptions. Cancelled subscriptions are displayed with **Inactive** status. Cancel subscription button will deactivate the customer token and no further charges will be make with it.

[Home](#) / [My Account](#)

Subscriptions

Subscription ID: 69c0aeb2-e34e-4d11-847a-6a217dbce435 Cancel	
	Last Order ID: 00050402 Next Charge Date: 10/5/23 Status: Inactive Total subscription amount: \$38.15
Subscription ID: 9404e832-2c91-4e8b-8535-6e26c881d9bf Cancel	
	Last Order ID: 00050403 Next Charge Date: 10/5/23 Status: Active Total subscription amount: \$38.15

Figure 37 My Account Subscription dashboard

2.3.19.5. *Recurring Subscription Order Creation*

A back-end job to process recurring subscription against each user in SFCC if subscription exists. The job iterates through customers and checks for subscriptions due for payment on the same day.

- Create a new order (with data from lastOrderID) and make a charge call to Klarna using the token and price from the lastOrderID.
- On success - update nextChargeDate based on subscriptionPeriod and set lastOrderID to the ID of the newly placed order. Update the same in the Subscription Dashboard against the subscription.
- On failure:
 - o Retry

Merchants can configure a retry mechanism where they can set a retry after specific number of days. If the retry fails, then the subscription can be Deactivated. This is generic configuration for retry:

- Retry: Boolean field with value Yes/No
- Number of Retry: 1,2 If Retry field is set as Yes, then merchant can configure this field.
- Retry Frequency: 1, 2 If Retry field is set as Yes, then merchant can configure this field.
- o Cancel subscription – if retry is disabled the subscription is cancelled.

Orders with trial period subscriptions are paid after the trial period is over. On the next charge date, a new order will be created.

New orders are with channel type = SUBSCRIPTIONS.

2.3.20. *Klarna Express Checkout*

Klarna Express Checkout is a new feature introduced in Storefront where Express Checkout Button will be displayed in PDP, Cart and Mini Cart and user will be redirected to Klarna after clicking Express Checkout button. Klarna Express Checkout enabled quick and easy checkout where Shipping Address, Billing Address and Payment details will be preselected so that Checkout can be completed in fewer clicks. Klarna Documentation - <https://docs.klarna.com/express-checkout/>. Multi step checkout is implemented with finalize call at place order stage.

2.3.20.1. *PDP*

- Button is not visible if the product is subscription only.

- Product is added to cart and the checkout is started with new basket only for this product. Once the checkout is completed the old basket is restored based on a session attribute.

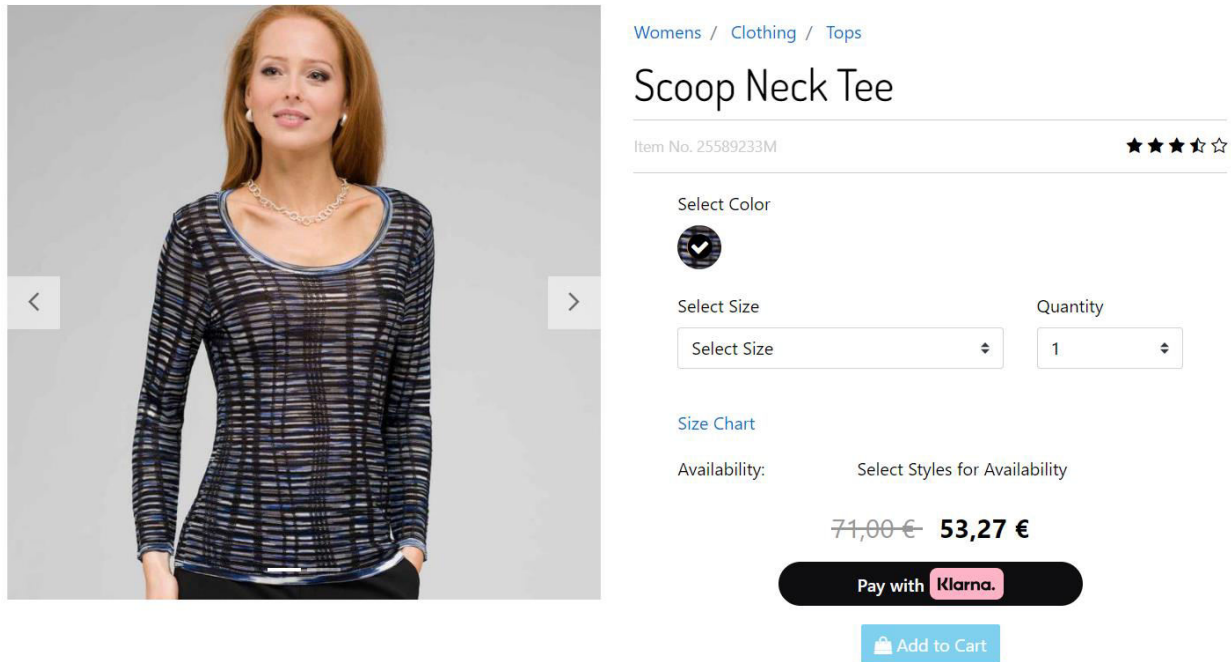


Figure 38 Klarna Express Checkout on PDP

2.3.20.2. Minicart and Cart buttons

- Express checkout button is not displayed in minicart if there are only subscription only products in the cart.

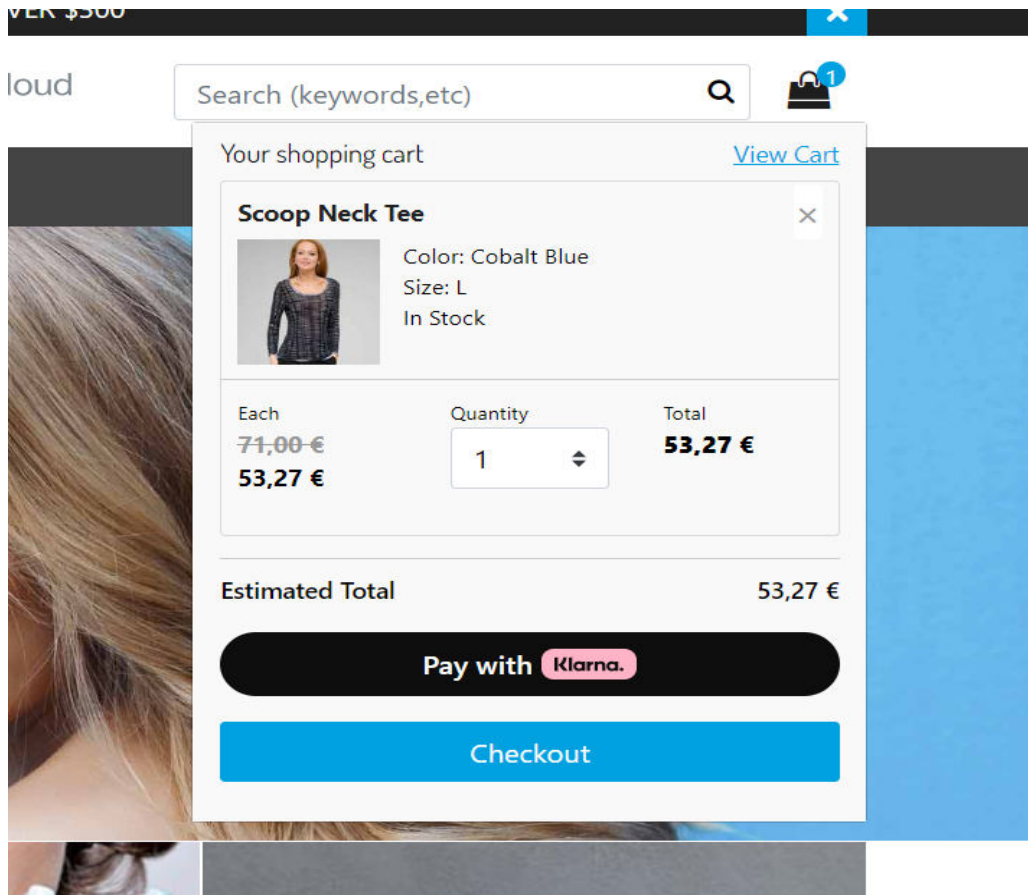


Figure 39 Klarna Express Checkout on minicart

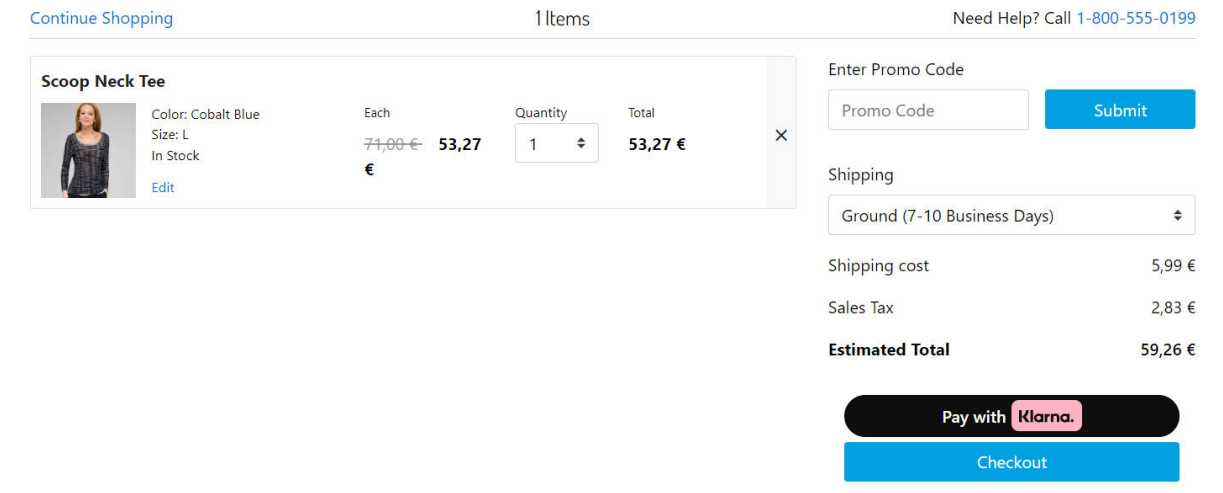


Figure 40 Klarna Express Checkout on cart page

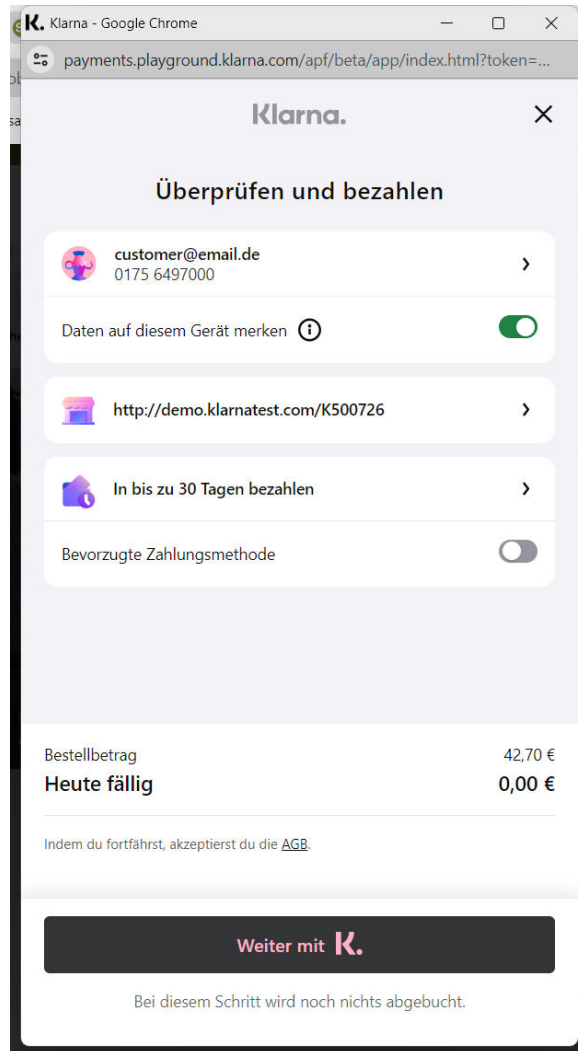


Figure 41 Klarna Express Checkout widget

2.3.20.3. Configuration (Deprecated)

Note: This configuration has been deprecated with release 24.4.0 and will be removed in the future.

- Basket has custom attribute “kpIsExpressCheckout” set as true for Express Checkout process.
- Merchant Tools > Site Preferences > Custom Site Preference Groups > Klarna Express Checkout

In Site Preferences we can enable/disable the feature and configure the buttons style. By default we have “Klarna” as payment method category which is displayed in checkout payment methods.

Name	Value	Default Value	
Klarna Express Checkout Enabled (kpECEnabled)	Yes	No	Edit Across Sites
Klarna Express Checkout Library URL (kpECLibraryURL) (String)	https://x.klarnacdn.net/kp/lib/v1/api.js	https://x.klarnacdn.net/kp/lib/v1/api.js	Edit Across Sites
Klarna Express Checkout Collect Shipping Address (kpECCollectShippingAddress)	Yes	No	Edit Across Sites
Klarna Express Checkout Button Theme (kpECButtonTheme)	Default (default)	Default	Edit Across Sites
Klarna Express Checkout Button Shape (kpECButtonShape)	Default (default)	Default	Edit Across Sites
Payment Category Content (kpECPaymentCategoryContent)	<pre>[{ "asset_urls": { "descriptive": "https://x.klarnacdn.net/payment-method/assets/badges/generic/klarna.svg", "standard": "https://x.klarnacdn.net/payment-method/assets/badges/generic/klarna.svg" }, "identifier": "klarna", "name": "klarna.payment.method" }]</pre>		Edit Across Sites
Placement (kec_placement)	None Cart (cart) pdp (pdp) Mini cart (minicart)		Edit Across Sites

Figure 42 Klarna Express Checkout configuration

- Merchant Tools > Custom Objects > Manage Custom Objects > KlarnaCountries
- Client key configuration by region in **“Klarna Express Checkout Client Key:”**

Minicart Express Button Shape:	- None -
Express Checkout	
Klarna Express Checkout Client Key:	klarna_test_client_VIU3KUNjMW53JDBVURUxucG1IZFlxmdNsc
<< Back to List	

Figure 43 KlarnaCountries Express Checkout Client Key

2.3.20.4. Configuration

- Basket has custom attribute “kpIsExpressCheckout” set as true for Express Checkout process.
- Merchant Tools > Site Preferences > Custom Site Preference Groups > Klarna Express Checkout

Merchant Tools / Site Preferences / Custom Site Preference Groups / Klarna Express Checkout

Cancel Apply to Other Sites Save

Instance Type
Sandbox

Search by ID

Name	Value	Default Value
Enable Express Checkout (kec_enable)	None	Yes
Theme (kec_theme)	None	Dark
Button Shape (kec_shape)	None	Rounded corners
Placement (kec_placement)	None Cart (cart) PDP (pdp) Mini cart (minicart)	

1-4 of 4

Edit Across Sites

Figure 44 Klarna Express Checkout configuration

2.3.21. Klarna SIWK (Sign in with Klarna)

Klarna SIWK is new feature introduced in where sign in with Klarna button will be displayed in login and checkout pages. User will get signed in through Klarna after clicking the siwk button. This will help users to easily checkout without entering the address as the user details will already be saved in the Klarna account.

2.3.21.1. Checkout page

salesforce commerce cloud

Checkout

Customer

Email

Continue as guest

Already have an account? Sign in

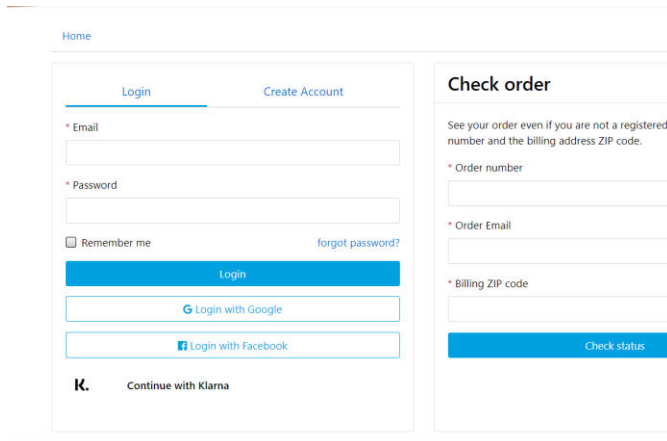
Login with Google

Login with Facebook

K. Continue with Klarna

After clicking on Continue with Klarna button, a popup will appear asking for country and contact information. User will get logged in and address fields will be populated so that user directly place order.

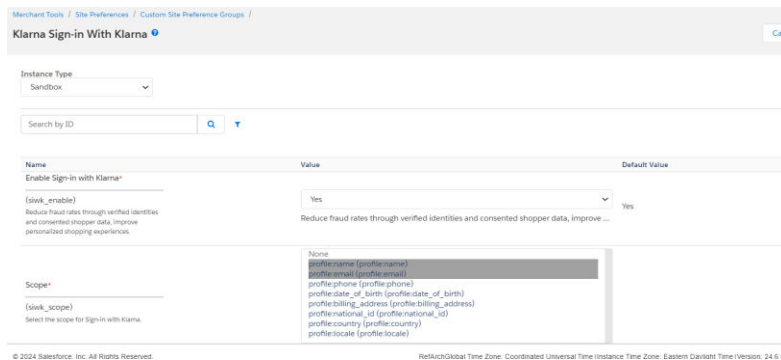
2.3.21.2. Login Page



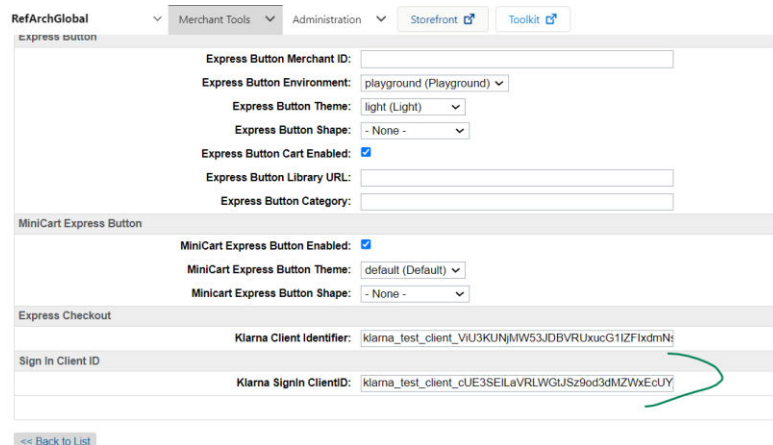
After clicking on Continue with Klarna button in login page, user will be redirected to account page .

2.3.21.3. Configuration

Klarna SIWK feature can be enabled in Site preferences under Klarna_siwk group preferences.



Client id is configured in Merchant Tools > Custom Object > Custom Object Editor.



Button styles can be changed in siwk_alignment, siwk_shape and siwk_theme in Site preferences. Button placement can be configured in login or checkout page as per site preference.

Note : Existing users will always be merged with Klarna account once signed in.

2.4. Compatibility

This cartridge has been tested against API Version: 22.6 (Compatibility Mode: 21.10) & SFRA version 6.0.0

2.5. Privacy, Payment

2.5.1. GDPR Compliance

The cartridge is compliant with GDPR recommendation and follows the best practice mentioned here and implementation transmits only required (PII) data to authorize payment method.

2.5.2. EMD (Extra Merchant Data)

The cartridge supports sending additional information on the customer's past purchase history, as well as “Buy Online, Pickup in Store” (BOPIS) store addresses when turned on in custom preferences: “Attachments” (~~kpAttachments (Deprecated with v24.4.0)~~ **kpEMD**). The type of data that can be send as an attachment is mentioned here. EMD is required for certain types of merchant orders and the inclusion of EMD is (e.g.customer_account_info: past interaction with merchant store) generally beneficial to improve acceptance rates.

EMD is included as part of Authorization step in Commerce Cloud checkout. The data send to Klarna is customizable and can be seen in “**int_klarna_payments/scripts/payments/additionalCustomerInfo.js**”. This script should return a JSON string to be used as a value for the body sub-field of the attachment field as [described here](#) .

If the example additionalCustomerInfo.js file is used unchanged the data send to Klarna is by the following schema:

```
{
  "$schema": "http://json-schema.org/draft-03/schema#",
  "id": "http://klarna.com/v2/emd#",
  "description": "Extended Merchant Data Payload Schema",
  "type": "object",
  "properties": {
    "customer_account_info": {
      "type": "array",
```

```

    "items": {
      "type": "object",
      "properties": {
        "unique_account_identifier": {
          "type": "string",
          "maxLength": 24
        },
        "account_registration_date": {
          "description": "ISO 8601 e.g. 2012-11-24T15:00",
          "type": "string",
          "format": "date-time",
          "pattern": "^[0-9][0-9][0-9][0-9]-[0-1][0-9]-[0-3][0-9]T[0-2][0-9]:[0-5][0-9](:[0-5][0-9]){0,1}Z{0,1}$"
        },
        "account_last_modified": {
          "description": "ISO 8601 e.g. 2012-11-24T15:00",
          "type": "string",
          "format": "date-time",
          "pattern": "^[0-9][0-9][0-9][0-9]-[0-1][0-9]-[0-3][0-9]T[0-2][0-9]:[0-5][0-9](:[0-5][0-9]){0,1}Z{0,1}$"
        }
      }
    },
    "payment_history_full": {
      "type": "array",
      "items": {
        "type": "object",
        "additionalProperties": false,
        "properties": {

```

```

    "unique_account_identifier": {
      "type": "string"
    },
    "payment_option": {
      "type": "string",
      "enum": ["card", "direct banking", "non klarna credit", "sms", "other"]
    },
    "number_paid_purchases": {
      "type": "integer"
    },
    "total_amount_paid_purchases": {
      "type": "number"
    },
    "date_of_last_paid_purchase": {
      "description": "ISO 8601 e.g. 2012-11-24T15:00",
      "type": "string",
      "format": "date-time",
      "pattern": "^[0-9][0-9][0-9][0-9]-[0-1][0-9]-[0-3][0-9]T[0-2][0-9]:[0-5][0-9](:[0-5][0-9]){0,1}Z{0,1}$"
    },
    "date_of_first_paid_purchase": {
      "description": "ISO 8601 e.g. 2012-11-24T15:00",
      "type": "string",
      "format": "date-time",
      "pattern": "^[0-9][0-9][0-9][0-9]-[0-1][0-9]-[0-3][0-9]T[0-2][0-9]:[0-5][0-9](:[0-5][0-9]){0,1}Z{0,1}$"
    }
  }
}

```

```

    "other_delivery_address": {
      "type": "array",
      "items": {
        "type": "object",
        "additionalProperties": false,
        "properties": {
          "shipping_method": {
            "type": "string",
            "enum": ["store pick-up", "pick-up point", "registered box", "unregistered
box"]
          },
          "shipping_type": {
            "type": "string",
            "enum": ["normal", "express"]
          },
          "first_name": {
            "type": "string"
          },
          "last_name": {
            "type": "string"
          },
          "street_address": {
            "type": "string"
          },
          "street_number": {
            "type": "string"
          },
          "postal_code": {
            "type": "string"

```



```

        "date_of_first_paid_purchase": "2015-03-18T20:03:03Z"
    },
    "other_delivery_address": [ {
        "shipping_method": "store pick-up",
        "shipping_type": "normal",
        "first_name": "Test",
        "last_name": "Customer",
        "street_address": "1487 Bay St",
        "street_number": "",
        "postal_code": "01109",
        "city": "Springfield",
        "country": "US"
    } ]
}
}
}
}

```

Please, note that in cases when the customer uses Guest Checkout, the EMD sent includes **payment_history_full[0].unique_account_identifier** (cqcid value set by SFCC), and all other fields are empty!!

2.5.3. PCI-DSS Compliance

Important Note: DO NOT SAVE UNENCRYPTED PCI DATA ON THE SERVER!

The virtual card (MCSv3) solution enables settlements using individual virtual card issued against a Klarna order. To be compliant with PCI-DSS requirements, merchant must ensure the data is securely maintained and transmitted as part of their operation in their live store environment. The required steps to ensure this, must be done in consultation with your payment service provider/acquirer and completed prior to go-live. Please review in advance the order

export details required for virtual card-based Klarna orders. Any historical decrypted PCI data should also be expunged, regardless of the VCN validity date.

3. Implementation Guide

3.1. Setup of Business Manager

The Klarna Payments LINK Cartridge contains 2 cartridges that are required for full functionality. Additionally, Controller and SFRA support is broken out into two separate cartridges, thereby facilitating the installation and use of one or the other models.

`int_klarna_payments` – Implements the core storefront functionality.

`int_klarna_payments_sfra` - Implements the storefront functionality with SFRA code.

3.1.1. Cartridge Upload & Assignment

Import the two cartridges into UX studio and associate them with a Server Connection.

- Import the “**int_klarna_payments**” cartridge into the SCC Studio Workspace:
 - Open SCC Studio
 - Click File -> Import -> General -> Existing Projects into Workspace
 - Browse to the directory where you saved the “**int_klarna_payments**” cartridge.
 - Click Finish.
 - Click OK when prompted to link the cartridge to the sandbox.
- Import the “**int_klarna_payments_sfra**” cartridge into the SCC Studio Workspace:
 - Open SCC Studio
 - Click File -> Import -> General -> Existing Projects into Workspace
 - Browse to the directory where you saved the “**int_klarna_payments_sfra**” cartridge.
 - Click Finish.
 - Click OK when prompted to link the cartridge to the sandbox.
- Prepend the Klarna cartridges to the effective site cartridge path:
 - Log into the SCC Business Manager.
 - Click Administration -> Sites -> Manage Sites.
- Select the desired site.

- Click on the Settings tab.
- Prepend “**int_klarna_payments_sfra:int_klarna_payments**” to the “**Cartridges**” field.
- Click Apply

The screenshot shows the 'Instance Type' dropdown set to 'Sandbox/Development'. Below it, a deprecation notice states: 'Deprecated. The preferred way of configuring HTTP and HTTPS hostnames is by using new features of the site aliases configuration ("SEO > Aliases Configuration"). The HTTP/HTTPS host is intended only to support an older configuration style.'

Fields for 'HTTP Hostname:' and 'HTTPS Hostname:' are present but empty.

Below the 'Instance Type: All' header, the 'Cartridges:' field contains the text: 'int_klarna_payments_sfra:int_klarna_payments:plugin_instorepickup:app_storefront'. The 'Effective Cartridge Path:' field lists the following components:

- int_klarna_payments_sfra
- int_klarna_payments
- plugin_instorepickup
- app_storefront_base
- modules
- plugin_apple_pay
- plugin_facebook
- plugin_payments
- plugin_pinterest_commerce
- plugin_web_payments
- bc_content
- core

Figure 45 Effective Cartridge Path

3.1.2. Metadata Import

- Go to main directory “**metadata**” folder, review the site-template content, and edit if needed. (Site template is prepared to setup “**SiteGenesis**” and “**RefArch**” sites - you may want to change that to your actual sites and delete the ones that are not needed). Zip the directory and you will have “**site-template.zip**” installation package.
- Log into the SCC Business Manager.
- Click Administration -> Sites Development -> Site Import & Export
- Browse to the directory where you saved the “**site-template.zip**”.
- Click “**Upload**”
- Select the uploaded site zip and click “**Import**”.

Note: Please, review the default **service.xml** file in the **site-template.zip** and update the configuration for Playground and Production accordingly before importing.

3.1.3. Build Klarna JS & CSS

If necessary, update the path to your base SFRA installation in **package.json** file from the same root folder.

Normally you would have a top-level project folder, into which the repositories of SFRA base cartridge and all required plugins, libraries and any other LINK cartridges will be cloned. In case you have cloned the Klarna cartridge into that folder as well, the below change will not be required. Otherwise, update **paths.base** property in the **package.json** to contain a relative path to the local directory containing the Storefront Reference Architecture repository. Its default value will be as follows:

```
"paths": {  
  "base": "../storefront-reference-architecture/cartridges/app_storefront_base/"  
}
```

Once you are certain the correct path to SFRA cartridge is configured, run “**npm run compile:scss**” command from the root folder of Klarna repository.

Run “**npm run compile:js**” command from the root folder of Klarna repository to build the Klarna specific JavaScript files.

3.2. Configuration

- Add your account settings to the KlarnaCountries Custom Objects.
 - Log into the SCC Business Manager.
 - Select the desired site from the tabs across the top of the page.
 - Click Custom Objects -> Custom Object Editor
 - Change the Object Type dropdown to “**KlarnaCountries**”.
 - Click the “**Find**” button.
 - Click the desired country you wish to edit (See screenshot below).
 - Update the required fields as mentioned in “KlarnaCountries” section
 - Repeat for the other countries.

Merchant Tools > Custom Objects > Custom Objects > US - General

General

Manage 'US' (KlarnaCountries)

Fields with a red asterisk (*) are mandatory. You can view and edit the name and description in other languages, if required. Click **Apply** to save the details.

custom	
Country Code:	US
On-site Messaging Data Default Locale:	en-US
Service Credential ID:	klarna.http.uscredentials
On-site messaging Data Client ID:	60a22a39-c2fd-5d09-bfe1-771459318a4d
Cart Placement Tag Enabled:	☒
Cart Placement Data Key:	info-page-standard
PDP Placement Tag Enabled:	☒
PDP Placement Data Key:	credit-promotion-small
Header Placement Tag Enabled:	☒
Header Placement Data Key:	top-strip-promotion-standard
Footer Placement Tag Enabled:	☒
Footer Placement Data Key:	footer-promotion-auto-size
Info Page Placement Tag Enabled:	☒
Info Page Placement Data Key:	info-page
Library URL:	https://na-library.playground.klarnaservices.com/lib.js

Figure 46 KlarnaCountries Settings

- Configure Klarna Payment Custom Preferences using the SCC Business Manager
 - Log into the SCC Business Manager
 - Select the desired site from the tabs across the top of the page.
 - Click Site Preferences -> Custom Preferences -> KlarnaPayment
 - Fill out the settings as desired. Descriptions of the site preferences are in the Site Preferences section.
- Configure Klarna Payment Service using the SCC Business Manager
 - Log into the SCC Business Manager
 - Click Administration > Operations > Services.
 - Click the Credentials tab.
 - Each Klarna credential correspond to one of the KlarnaCountries custom objects. Click on the one you want to edit.
 - Enter the MID API username and API password you received from Klarna.

- Edit URL field if Production environment. Klarna API URLs information - <https://developers.klarna.com/api/#api-urls>.

[Administration](#) > [Operations](#) > [Services](#) > [Service Credentials](#) > klarna.http.gbcredentials - Details

klarna.http.gbcredentials

Fields with a red asterisk (*) are mandatory. Click **Apply** to save the details. Click **Reset** to revert to the last saved state.

These credentials are used by 0 services.

Name: *	klarna.http.gbcredentials
URL:	https://api.playground.klarna.com/
User:	your Merchant ID
Password:	••••••••

Figure 47 Service Settings

- Configure Klarna Rate Limited Service Profile using the SCC Business Manager (Optional)
 - Log into the SCC Business Manager
 - Navigation : Merchant Tools > Site preferences > Custom Site Preferences group > Klarna Payments
 - Enable Service Limit configuration > Select “Rate Limit By Operation” to Yes. If it is selected to NO, the default service will control the rate limit.
- Configure Custom Rate limits in SCC Business Manager (Optional)
 - Log into the SCC Business Manager
 - Navigation: Administration > Operations > Services
 - Select a required profile (e.g : Klarna.http.createSession)
 - Enable Rate Limit: Check box
 - Max Rate Limit Calls: 50 (e.g.: Higher rate limit of 50 requests/sec agreed with Klarna)
 - Rate Limit Internal (millisecond): 1000
 - Repeat for all Service operation with respective agreed rate limits



Sandbox - zzjh
RefArch ▼

Merchant Tools ▼

Administration ▼

Storefront

Toolkit

Administration > Operations > Services > Service Profiles > klarna.http.createSession - Details

klarna.http.createSession

Fields with a red asterisk (*) are mandatory. Click **Apply** to save the details. Click **Reset** to revert to the last saved state.

This profile is used by 1 service.

Name: *

klarna.http.createSession

Connection Timeout (ms):

30,000

Enable Circuit Breaker:

☐

Max Circuit Breaker Calls:

0

Circuit Breaker Interval (ms):

0

Enable Rate Limit:

☒

Max Rate Limit Calls:

50

Rate Limit Interval (ms):

1000

[<< Back to List](#)

3.3. Extended Controllers

Controller	Start Node	Remarks
Checkout.js	Begin	Extended to call Klarna session manager
CheckoutServices.js	Get, SubmitPayment, PlaceOrder	Klarna payment method/category and totals are being stored
CheckoutShippingServices.js	SubmitShipping, ToggleMultiShipping	Calling the Klarna session manager
Order.js	Confirm	Extending Klarna order data to view data

Table 1 Extended Controllers List

3.4. Template Updates

Templates have been updated to support On-site messaging and Addresses forms for Klarna. To be used as reference but feel free to customize the templates to match your specific needs. Final review and sign-off as per project requirements and contract agreements.

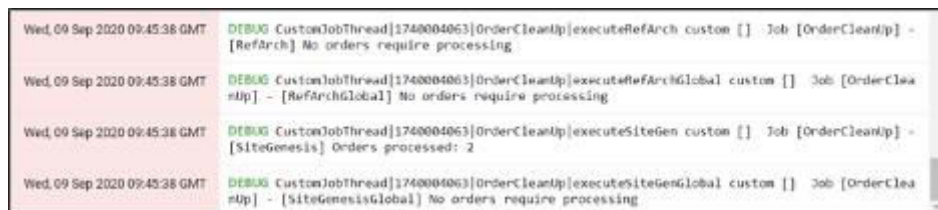
3.5. Jobs

3.5.1. Job “OrderCleanUp” (Optional)

This 1-time clean-up job is only applicable to merchants integrated with Klarna Payments cartridge version (< 19.1.6), utilizing (or previously used) virtual card-based settlement (VCN) and stored decrypted card details within Business Manager.

The job iterates over orders with status “**Exported**” and attribute “**custom.kpIsVCN=true**” to remove the sensitive details saved in fields kpVCNPAN, kpVCNCSC, kpVCNExpirationMonth, kpVCNExpirationYear as part of the previous releases. There are no parameters passed to the script.

Upon successful run, the job will log the result of processed orders in the custom debug log located in “**webdav/Sites/Logs**”. Depending on the setup, you will receive a message for the processed orders count for each storefront or message that there are no orders needing update.



```
Wed, 09 Sep 2020 09:45:38 GMT  DEBUG CustomJobThread[1740004063]OrderCleanUp|executeRefArch custom {} Job [OrderCleanUp] - [RefArch] No orders require processing
Wed, 09 Sep 2020 09:45:38 GMT  DEBUG CustomJobThread[1740004063]OrderCleanUp|executeRefArchGlobal custom {} Job [OrderCleanUp] - [RefArchGlobal] No orders require processing
Wed, 09 Sep 2020 09:45:38 GMT  DEBUG CustomJobThread[1740004063]OrderCleanUp|executeSiteGen custom {} Job [OrderCleanUp] - [SiteGenesis] Orders processed: 2
Wed, 09 Sep 2020 09:45:38 GMT  DEBUG CustomJobThread[1740004063]OrderCleanUp|executeSiteGenGlobal custom {} Job [OrderCleanUp] - [SiteGenesisGlobal] No orders require processing
```

Figure 48 Job Logs

Upon error, the cause of the failure (message and stack) will be logged in the standard error log.



```
Wed, 09 Sep 2020 09:05:22 GMT  ERROR CustomJobThread[618064818]OrderCleanUp|execute com.demandware.beehive.core.internal.do
main.SystemObjectQueryImpl Sites-RefArch-Site JOB 43e0d91e96 e67f8cb0ade56d99ae0c4bc12b
1648624161284573384 - Exception while parsing system object query:
System
RequestID: e67f8cb0ade56d99ae0c4bc12b
SessionType: JOB
Truncated SessionID: 43e0d91e96
User Profile UUID: 03e37ff77529fda91ae75408ab
```

Figure 49 Job Logs

To setup the job, go to **Administration > Operations > Import & Export** and import file “**jobs.xml**”. Out of the box, the XML file includes only the RefArch scope, but it can be configured with multiple flows if you have more than one site using this functionality as seen bellow. Each site should be added as a separate flow.

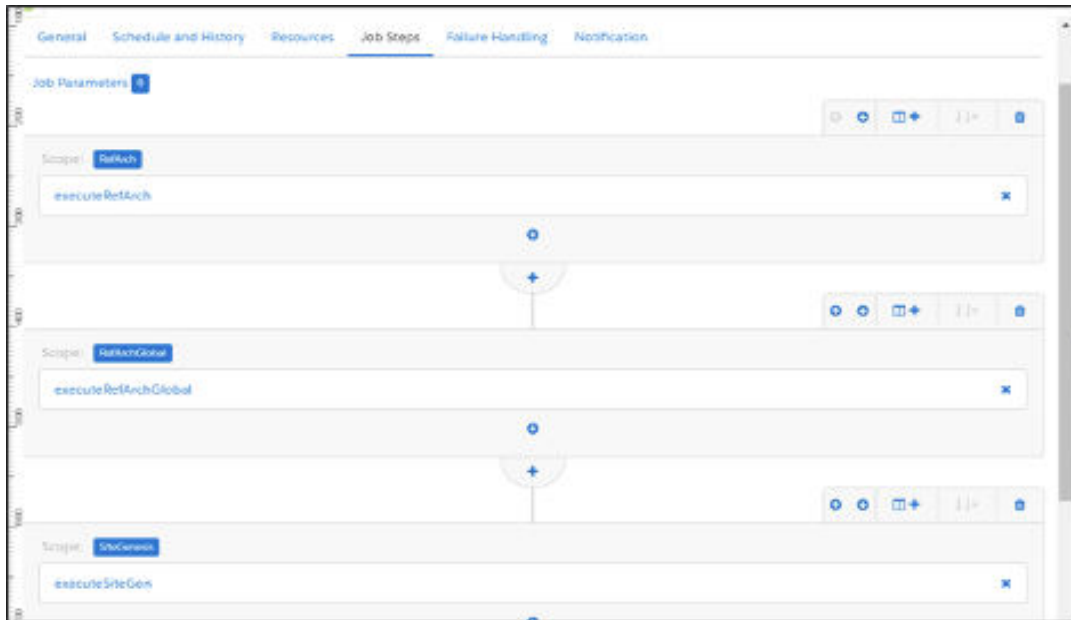


Figure 50 Job Steps

To set up the required job parameters and add new flow, follow the bellow steps. If you only have one storefront and need to change the scope to the correct one – proceed to steps 4-5 directly.

1. Click on the “Add a sequential flow” button at the bottom of the current flow.



Figure 51 Add New Job Step

2. Once done, click on “Configure a step” button within the flow. In the flyout that has just opened search for “script” and select “ExecuteScriptModule”.

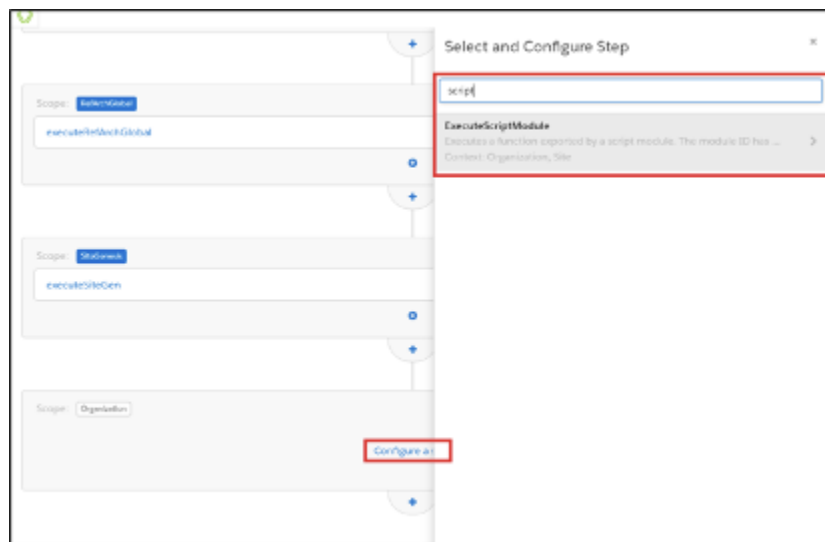


Figure 52 Configure Step

3. In the flyout populate these fields and click the “Assign” button.
 - a. ID – Enter any meaningful name in the field. In case you have multiple flows, this should not be a duplicate one. If you enter a duplicate name, the details won’t be saved and you will see an error message.
 - b. ExecuteScriptModule.Module – Enter the location of the “OrderCleanUpJob.js” file. Out of the box it should be

“int_klarna_payments/cartridge/scripts/job/OrderCleanUpJob.js” or the location where you have placed the script file.

- c. ExecuteScriptModule.FunctionName – Leave the field value to “execute”

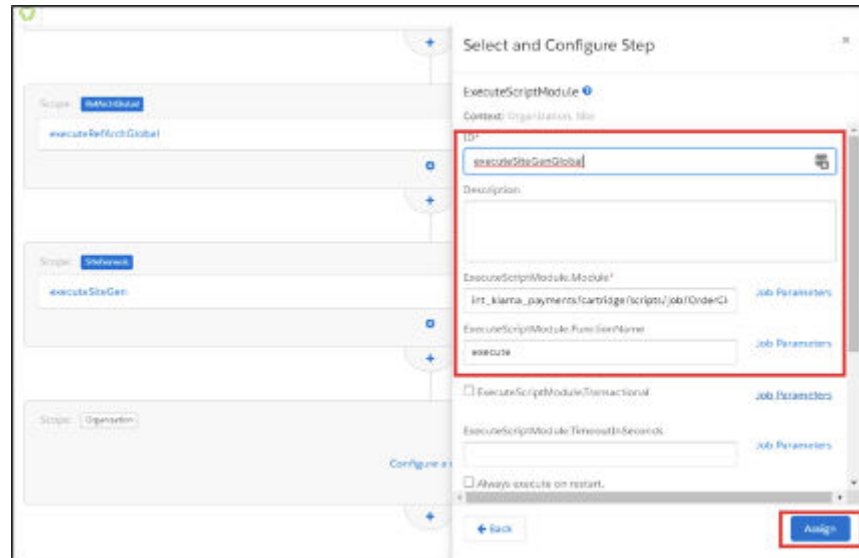


Figure 53 Configure Step (cont.)

- 4. Once the step has been added, you should make sure to assign it to the correct site scope. Click on “Organization” and select “Specific Sites” from the drop-down.

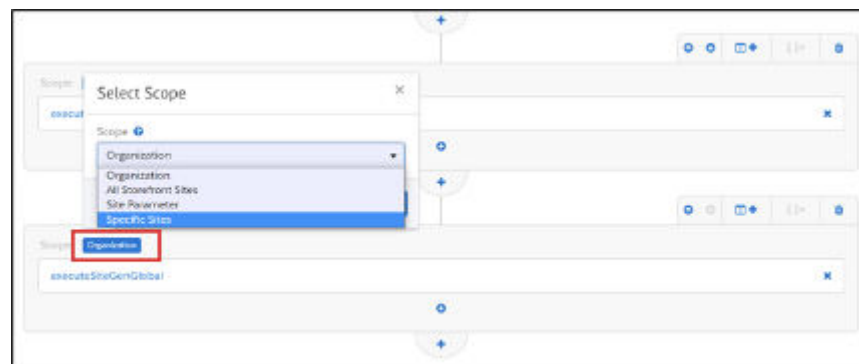


Figure 54 Job Scope

- From the list of sites, select the respective site ID (i.e. SiteGenesisGlobal) and click on “Assign”.

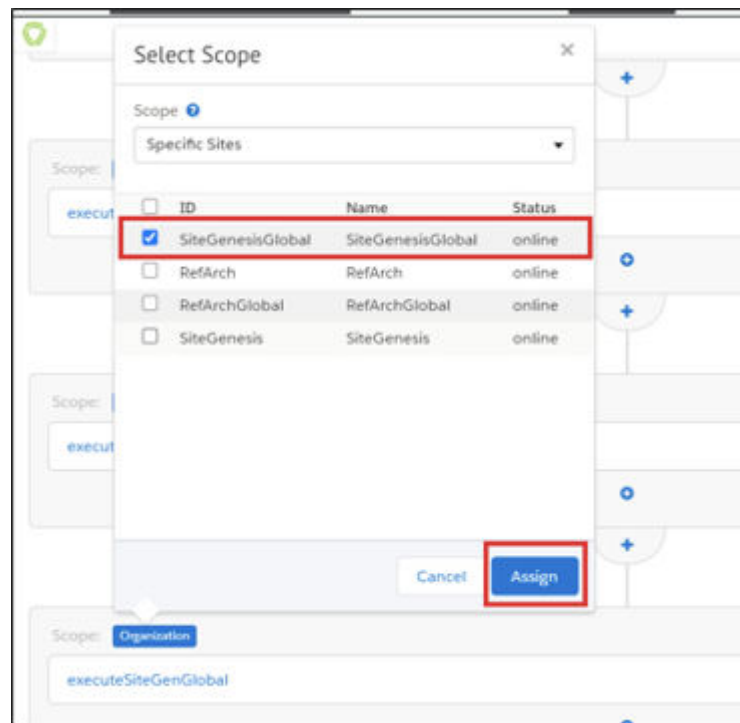


Figure 55 Job Scope (cont.)

- Repeat steps 1-5 for each site/storefront that you have using Klarna VCN and nned additional configuration.

3.5.2. Job “RecurringOrders”

The job iterates over all customers checking for subscriptions entries. Process all subscription entries eligible for charge – subscription should be enabled and nextChargeDate or nextRetryDate should match the current date. New SFCC orders are created and the old ones are replaced. Orders with expiring trial period, are only charged.

By default in jobs.xml the job is configured to run on RefArch site but this can be changed either in file or in storefront.

The job has one job step – createOrder with following configuration:

- ExecuteScriptModule.Module - **int_klarna_payments/cartridge/scripts/job/RecurringOrdersJob.js**
- ExecuteScriptModule.FunctionName – **execute**

The job is executed on site level.

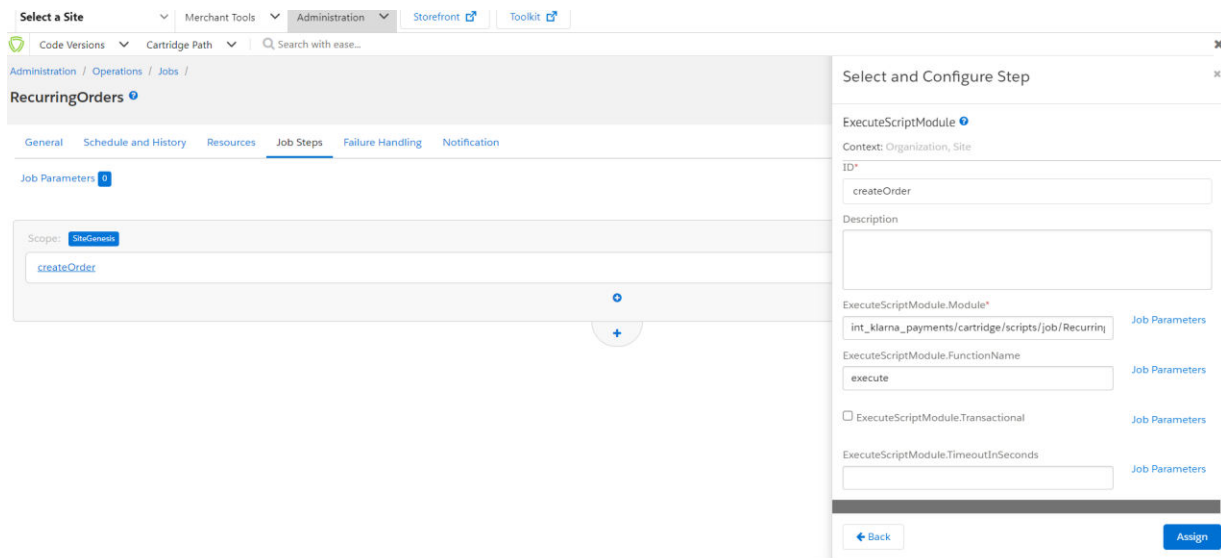


Figure 56 RecurringOrdersJobs

3.6. Custom Code

Store Front Reference Architecture (SFRA) does not imply modification of core cartridge to enable any of LINK integration cartridges. The "int_klarna_payments_sfra" cartridge follows best practices recommended by Salesforce. Refer to below 2 cases where changes may be applicable:

3.6.1. *app_storefront_base\cartridge\client\default\js\checkout\checkout.js (optional)*

After placing an order every customer is redirected to Klarna and then sent back to the site with order confirmation page. To prevent sending of any additional URL params to Klarna please do the following:

<pre> 310 else if (stage === 'placeOrder') { 311 // disable the placeOrder button here 312 \$('body').trigger('checkout:disableButton', '.next-step-button button'); 313 \$.ajax({ 314 url: \$('.place-order').data('action'), 315 method: 'POST', 316 success: function (data) { 317 // enable the placeOrder button here 318 \$('body').trigger('checkout:enableButton', '.next-step-button button'); 319 if (data.error) { 320 if (data.cartError) { 321 window.location.href = data.redirectUrl; 322 defer.reject(); 323 } else { 324 // go to appropriate stage and display error message 325 defer.reject(data); 326 } 327 } else { 328 var continueUrl = data.continueUrl; 329 var urlParams = { 330 ID: data.orderID, 331 token: data.orderToken 332 }; 333 } 334 continueUrl += (continueUrl.indexOf('?') !== -1 ? '&' : '?') + 335 Object.keys(urlParams).map(function (key) { 336 return key + '=' + encodeURIComponent(urlParams[key]); 337 }).join('&'); </pre>	<pre> 310 else if (stage === 'placeOrder') { 311 // disable the placeOrder button here 312 \$('body').trigger('checkout:disableButton', '.next-step-button button'); 313 \$.ajax({ 314 url: \$('.place-order').data('action'), 315 method: 'POST', 316 success: function (data) { 317 // enable the placeOrder button here 318 \$('body').trigger('checkout:enableButton', '.next-step-button button'); 319 if (data.error) { 320 if (data.cartError) { 321 window.location.href = data.redirectUrl; 322 defer.reject(); 323 } else { 324 // go to appropriate stage and display error message 325 defer.reject(data); 326 } 327 } else { 328 var continueUrl = data.continueUrl; 329 var urlParams = {}; 330 if (data.orderID && data.orderToken) { 331 urlParams.ID = data.orderID; 332 urlParams.token = data.orderToken; 333 } 334 continueUrl += (continueUrl.indexOf('?') !== -1 ? '&' : '?') + 335 Object.keys(urlParams).map(function (key) { 336 return key + '=' + encodeURIComponent(urlParams[key]); 337 }).join('&'); </pre>
--	---

Figure 57 Changes in checkout.js

3.6.2. *app_storefront_base\cartridge\client\default\js\cart\cart.js (required)*

In order get updates of the cart OSM widget please do the following in cart.js or in your app cartridge.

In function `updateCartTotals(data) {}` add the following code at the bottom:

```

if (data.totals.klarnaTotal) {
  $('klarna-placement').attr('data-purchase-amount', data.totals.klarnaTotal);
  if (window.Klarna && window.Klarna.OnsiteMessaging) {
    window.Klarna.OnsiteMessaging.refresh();
  }
}

```

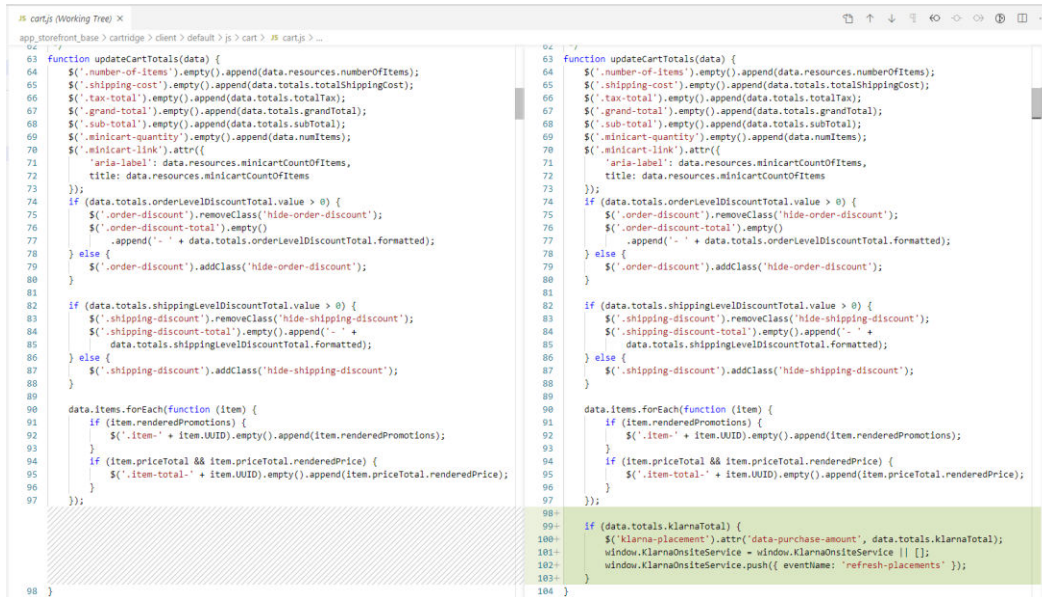


Figure 58 Changes in cart.js

3.7. Integration with Other Payment Cartridges

The contents of this section are only applicable if there are other payment cartridge integrations within the cartridge path besides **Klarna Payments SFRA**. No matter of the order of those cartridges in the cartridge path, there are certain templates that need to be overwritten by adding a new if-condition and including the right sub-template:

- \templates\default\checkout\billing\paymentOptions\paymentOptionsContent.isml
- \templates\default\checkout\billing\paymentOptions\paymentOptionsSummary.isml
- \templates\default\checkout\billing\paymentOptions\paymentOptionsTabs.isml

Suppose the website owner requires PayPal as well as Klarna. Each of these templates mentioned above must be copied to a new custom cartridge. The example below shows the new code of the *paymentOptionsContent.isml* template:

```
<isloop items="{pdict.order.billing.payment.applicablePaymentMethods}"
var="paymentOption" status="loopSate">

  <isif condition="{paymentOption.ID === 'CREDIT_CARD'}">

    <isinclude
template="checkout/billing/paymentOptions/creditCardContent" />

  </isif>

  <isif condition="{paymentOption.ID === 'KLARNA_PAYMENTS'}">
```

```
<isinclude  
template="checkout/billing/paymentOptions/klarnaPaymentsContent" />  
  
</isif>  
  
...  
  
<isif condition="{paymentOption.ID === 'PayPal'}">  
    <isinclude template="paypal/checkout/paypalContent" />  
</isif>  
</isloop>
```

Same goes for the other two templates.

3.8. External Interfaces

All requests are done through Klarna's REST API and encrypted using SHA-256 with the shared secret provided by Klarna. Only HTTPS is allowed. JSON is used across all communications.

The full reference guide, along with the resource structure for requests & responses, can be found in the developer portal - <https://docs.klarna.com/klarna-payments/api/>

4. Testing

Klarna has a set of testing credentials and triggers that can be used.

Please, refer to the following URL: <https://docs.klarna.com/resources/test-environment/>

5. Operations, Maintenance

5.1. Data Storage

5.1.1. System Object Extensions

5.1.1.1. Basket

Parameter Name	Attribute ID	Description
Klarna Session ID	kpSessionId	The Klarna session ID returned after “Create Session” API endpoint is called <i>(Applicable since version 21.2.0)</i>
Klarna Client Token	kpClientToken	Client token returned by “Create Session” API endpoint and used to initialize the JS SDK <i>(Applicable since version 21.2.0)</i>
Klarna Client Token	kpClientToken	Client token returned by “Create Session” API endpoint and used to initialize the JS SDK.
Klarna Subscription Frequency	kpSubscriptionFrequency	Subscription frequency values (day, month, etc.)
Klarna Subscription Period	kpSubscriptionPeriod	Predefined subscription period in numbers
Klarna Is Express Checkout	kpIsExpressCheckout	Determines if the basket is an express checkout

Table 2 Basket Attributes

5.1.1.2. Order

Parameter Name	Attribute ID	Description
Klarna Payments Order ID	kpOrderID	The Klarna payments Order ID for Klarna payment method selected by customer

VCN Brand	kpVCNBrand	Klarna Payments virtual card scheme name
VCN Holder	kpVCNHolder	Klarna Payments virtual card holder name
VCN Card ID	kpVCNCardID	Klarna Payments Virtual Card - Card ID
VCN PCI Data	kpVCNPCIData	Klarna Payments Virtual Card PCI Data in encrypted format
VCN Initialization Vector	kpVCNIV	Klarna Payments Virtual Card Initialization Vector
VCN AES Key	kpVCNAESKey	Klarna Payments Virtual Card AES Key
Is VCN Used	kpIsVCN	True if virtual card is enabled & used for payment of the order, otherwise false
Klarna Session ID	kpSessionId	The Klarna session ID returned after “Create Session” API endpoint is called <i>(Applicable since version 21.2.0)</i>
Klarna Client Token	kpClientToken	Client token returned by “Create Session” API endpoint and used to initialize the JS SDK <i>(Applicable since version 21.2.0)</i>
Klarna Subscription Frequency	kpSubscriptionFrequency	Subscription frequency values (day, month, etc.)
Klarna Subscription Period	kpSubscriptionPeriod	Predefined subscription period in numbers

Table 3 Order Attributes

5.1.1.3. Order Payment Instrument

Parameter Name	Attribute ID	Description
Klarna Payment Category ID	klarnaPaymentCategoryID	ID of Klarna payment category
Klarna Payment Category Name	klarnaPaymentCategoryName	Name of Klarna payment category

Table 4 Order Payment Instrument Attributes

5.1.1.4. Payment Transaction

Parameter Name	Attribute ID	Description
Fraud Status	kpFraudStatus	Klarna Payments order fraud status
Klarna Authorization Token	kpAuthorizationToken	This attribute stores the Klarna authorization token, which is a string value used to authenticate and finalize the order creation process with Klarna's payment system. (Applicable since version 23.2.0)
Klarna Redirect URL	kpRedirectURL	This attribute holds the URL to which the customer is redirected after the payment authorization is successfully completed by Klarna's system. (Applicable since version 23.2.0)

Table 5 Payment Transaction Attributes

5.1.1.5. Site Preferences

The site custom preferences have been extended with a new group called “*Klarna_Payments*”. The table below describes the preferences within that group:

Parameter Name	Attribute ID	Description
Auto-capture	kpAutoCapture	When enabled “Yes”, a full order capture will be attempted automatically. The standalone

		order management API capture request will include total order amount value for “captured_amount”. Default value is “No”
Klarna Payments Service Name	kpServiceName	The service name used for the current site
Send product_url and image_url	sendProductAndImageURLs	If set to true, product_url and image_url fields will be included in the Klarna session and order API calls. This enhances shopper experience post purchase. Default value is “Yes”
Merchant Reference 2 Mapping	merchant_reference2_mapping Deprecated and renamed to merchant_reference2 under Klarna_KP	The field from SCC order (basket) object that is mapped to merchant_reference2 field from klarna API request. Has to be one of the class attributes of SCC LineItemCtnr. Note that for complex data structures result may vary. Note: Merchant Reference 1 value is always set to the SCC order ID
Border Color Preference	kpColorBorder Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for Border in Klarna Payments iFrame
Border Selected Color Preference	kpColorBorderSelected Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for selected element Border in Klarna Payments iFrame

Button Color Preference	kpColorButton Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for Button in Klarna Payments iFrame
Button Text Color Preference	kpColorButtonText Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for Button text in Klarna Payments iFrame
Checkbox Color Preference	kpColorCheckbox Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for Checkbox in Klarna Payments iFrame
Checkbox Checkmark Color Preference	kpColorCheckboxCheckmark Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for checkbox checked(selected) in Klarna Payments iFrame
Details Color Preference	kpColorDetails Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for details in Klarna Payments iFrame
Header Color Preference	kpColorHeader Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for Header in Klarna Payments iFrame
Rate limit By Operation	kpRateLimitByOperation	Select “Rate Limit By Operation” to Yes. If it is selected to NO, the default service profile will be utilized. The standard API rate limit for the Klarna service is as mentioned listed on docs.klarna.com . The default service id is klarna.http.defaultendpoint.

Klarna Payment Create New Session When Expires	kpCreateNewSessionWhenExpires	If set to Yes, then a new Klarna session will be created if Klarna session expires before SFCC basket expires
Link Color Preference	kpColorLink Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for link in Klarna Payments iFrame
Text Color Preference	kpColorText Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for text in Klarna Payments iFrame
Secondary Text Color Preference	kpColorTextSecondary Deprecated and used in kpColorCustomization under site preference group Klarna_KP	CSS (hex value) color set for secondary text in Klarna Payments iFrame
Border Radius Preference	kpRadiusBorder Deprecated and used in kpColorCustomization under site preference group Klarna_KP	Value (in pixels) of the border radius to be used in Klarna Payments iFrame
Attachments	kpAttachments Deprecated and renamed to kpEMD under Klarna_KP	Toggle (Yes/No) for the inclusion of attachments when creating an order. Specific to inclusion of EMD (customer_account_info, other_delivery_address) when applicable. Default is “No”.
Not available message on billing page	kpNotAvailableMessage	The Klarna Payment not available message on billing page. JSON string holding country code and corresponding message string. For example:

		<pre>{ "GB": "Klarna Payment not available", "default": "Klarna Payment not available" }</pre> Note: This is deprecated and will be removed in next releases!
Virtual Card Number Enabled	kpVCNEnabled	If this option is set to “Yes”, Klarna settlement request will generate a Virtual Card Number for every Klarna order. Note: the option will only work if VCN private/public keys are configured properly as mentioned below and public key shared in advance with Klarna
VCN Public Key ID	kpVCNkeyId	UUIDv4 value corresponding to the key pair. Shared with Klarna representative for Production & Playground (test) env configuration
VCN Private Key	vcnPrivateKey	SSL private key used only to decode Virtual Card information (used with kpVCNEnabled). Refer to section 9.3 Decrypt VCN Card Details
VCN Public Key	vcnPublicKey	SSL public key used with Virtual Card integration (used with kpVCNEnabled). Shared with Klarna and stored here for reference.
VCN Settlement Retry Enabled	kpVCNRetryEnabled	If set to “Yes”, SFCC will retry the VCN settlement once again in

	Deprecated and Renamed to kpVCNRetry under group Klarna_KP	case of service error. Default is “No”
Promotion Price Taxation	kpPromoTaxation	Only use "Based on Adjusted Price" value if you have enabled the corresponding value in “Merchant Tools > Site Preferences > Promotions > Discount Taxation” and use gross taxation. Default: Based on Price
Hide Payment Methods on Deny	kpRejectedMethodDisplay	If set to value other than "No", the Klarna payment method options on the checkout will be greyed out or not displayed to customer in the current view when Klarna authorization request is rejected in the response (.i.e hard reject - "show_form" and "approved" values are both "false")
Alternative Klarna Payment Flow	kpUseAlternativePaymentFlow	If set to "Yes", Klarna Authorization and Order creation steps will be triggered on Checkout Review page when customer clicks CTA/Place Order Button. Default: No
Enable OMS	kpOMSEnabled	If set to "Yes", it updates the order information ingested in SFOMS (SalesForce Order Management System) in the format that Klarna expects. It is required when there is an

		integration between SFCC and SFOMS. Default: No
Klarna Payments Additional Logging	kpAdditionalLogging Deprecatcd and Renamed to kp_log_data under Klarna_KP	If set to "Yes", it writes additional logging info to get more order details when issue occurs. Default: No
Agent User Name	kpAgentUserName Deprecatcd and renamed to kpSubsUsername under Klarna_KP	User Agent Name for orders on behalf
Agent User Password	kpAgentUserPassword Deprecatcd and renamed to kpSubsPassword under Klarna_KP	User Agent Password for orders on behalf
Enable Retry for Recurring Orders	kpEnableRecurringOrderRetry Deprecatcd and renamed to kpSubsRetryEnable under Klarna_KP	Enable retry for failed orders
Number Of Retries	kpRecurringNumberOfRetry Deprecatcd and renamed to kpSubsRetryNumber under Klarna_KP	Recurring number of retries
Recurring Retry Frequency	kpRecurringRetryFrequency Deprecatcd and renamed to kpSubsRetryFrequency under Klarna_KP	Retry frequency for recurring orders
Klarna Create Order Token	kpCreateOrderToken	Token used to verify the caller for recurring orders

Use Bank Transfer callback	kpBankTransferCallback	If set to “Yes”, SFCC will wait for Klarna callback to place an order. Default is “No”
Klarna Express Checkout (Klarna_ExpressCheckout) deprecated from 24.4.0		
Klarna Express Checkout Enabled	kpECEEnabled	Boolean flag to enable/disable Klarna Express Checkout
Klarna Express Checkout Button Theme	kpECButtonTheme	The theme of the button. Options include default, light & dark
Klarna Express Checkout Button Shape	kpECButtonShape	The shape of the button. Options include default, rect & pill
Placement	kec_placement	Multiselect attribute to choose where to display express checkout buttons. Values: cart, pdp, minicart
Klarna Express Checkout (Klarna_KEC)		
Enable Express Checkout	kec_enable	Boolean flag to enable/disable Klarna Express Checkout
Theme	kec_theme	The theme of the button. Options include default, light & dark
Button Shape	kec_shape	The shape of the button. Options include default, rect & pill
Placement	kec_placement	Multiselect attribute to choose where to display express checkout buttons. Values: cart, pdp, minicart
Use Klarna One-Step Checkout	kec_useOneStepCheckout	Use Klarna One-Step Checkout
Klarna_SignIn(Klarna Sign In)		
Enable/Disable Klarna Sign In	isKpSignInEnabled – Deprecated with version 24.4.0	Enable /Disable Klarna sign in

	Renamed to siwk_enable	
Klarna Button Logo Alignment	kpSignInButtonLogoAlignment - Deprecated with version 24.4.0 Renamed to siwk_alignment	Alignment of Klarna Button.Options include Badge (Recommended), Left, Center. Default value is Badge(Recommended)
Klarna Button Shape	kpSignInButtonShape - Deprecated with version 24.4.0 Renamed to siwk_shape	Shape of Klarna Button.Options include Rounded corners (Recommended), Rectangular, Pill with default value Rounded corners(Recommended)s
Klarna Button Theme	kpSignInButtonTheme - Deprecated with version 24.4.0 Renamed to siwk_theme	Theme of Klarna Button.Options include Light (Recommended), Dark, Outlined with default value Light(Recommended)
Customer Unique Identifier Login Method	kpSignInCustomerUniqueId - Deprecated with version 24.4.0	Options include Bank ID and OTP
Klarna SignIn Environment	kpSignInEnvironment - Deprecated with version 24.4.0 Removed from preference and handled in code with value playground	Preference for configuring environment.Options include Playground and Production
Klarna Sign in web flow options	kpSignInInteractionMode - Deprecated with version 24.4.0 Removed from preferences and used default value as DEVICE_BEST	Interaction mode when Klarna button is clicked .Options include redirect url,popup and device best.Default value is device best
Auto Login to Klarna payments	kpSignInIsAutoLogin - Deprecated with version 24.4.0	Boolean to automatically loginto Klarna payments

Link Existing user	kpSignInLinkExistingUser - Deprecated with version 24.4.0 Always merge accounts without any prmpt	To link existing users.Options include Merge accounts automatically and prompt to merge accounts
Redirect Uri	kpSignInRedirectUri - Deprecated with version 24.4.0 Renamed to siwk_redirect_url	Redirect url after sign-in
Script URL	kpSignInScriptURL - Deprecated with version 24.4.0 Removed prefereces and handled in code .Used the script url in constants	Klarna Sign in Script URL
Sign In scope	kpSignInScope - Deprecated with version 24.4.0 Renamed to siwk_scope	Sign in scope.Uses a multiselect list of all scopes available
Sign In Placement	siwk_placement	Placement of sign in button.Multiselect List of strings which include loginpage and checkout page.
Klarna OSM(On site Messaging)		
Enable On-Site Messaging	osm_enable	Enable/Disable OSM functionality
Theme	osm_theme	Theme for on site messaging .Options include default,dark and custom
Placement	osm_placement	Multiselect dropsown which includes cart,pdp,header,footer and info. Placement Tag IDs are hardcoded in code
Custom Styling	osm_custom_styling	JSON to store custom styling

Klarna_KP(Klarna Payments)		
Enable Klarna Payments	kp_enable	Enable/Disable Klarna Payments. Boolean attribute with Default value as true
Color Customization	kpColorCustomization	JSON attribute to customize colour
Enable Extra Merchant Data	kpEMD	Enable this option to include customer_account_info and other_delivery_address as attachments when creating an order. Boolean with default value false
VCN - Enable settlement retry	kpVCNRetry	When set to "Yes", SFCC will automatically retry the VCN settlement in the event of a service error. Boolean with default value false
Log Extra debug data	kpLogExtraData	Attribute to log debug data. Boolean with default value true
Merchant Reference 2	merchant_reference2	Enter the attribute from the SCC order (basket) object that you would like to forward as an additional merchant reference to Klarna. This attribute must be a class attribute of the SCC LineItemCtnr. String attribute
Agent username for subscriptions	kpSubsUsername	Agent username used to login on behalf of customers to create subscription orders. String attribute
Agent password for subscriptions	kpSubsPassword	Agent password to login on behalf of customers to create

		subscription orders.Password Attribute
Enable retry for subscriptions	kpSubsRetryEnable	Boolean with default value false
Number of Retries	kpSubsRetryNumber	Number attribute
Recurring retry frequency	kpSubsRetryFrequency	Number attribute

Table 6 Site Preferences

5.1.1.6. *Product*

Parameter Name	Attribute ID	Description
Is Klarna Standard Product	kpIsStandardProduct	Boolean attribute to define if the product is standard. <i>(Applicable since version)</i>
Is Klarna Subscription Product	kpIsSubscriptionProduct	Boolean attribute to define if the product is eligible for subscription.
Klarna Trial Days Usage	kpTrialDaysUsage	Numeric value used for free trial definition.

5.1.1.7. *ProductLineItem*

Parameter Name	Attribute ID	Description
Is Selected for Subscription product	kpSubscription	Boolean attribute to define if the product is selected for subscription in the basket.
Enriched data for product line item.	klarna_oms__lineItemJSON	String data for product line item.

5.1.1.8. *Profile*

Parameter Name	Attribute ID	Description
----------------	--------------	-------------

Klarna Subscriptions	kpSubscriptions	Text attribute to store customer subscriptions to Klarna
----------------------	-----------------	--

5.1.2. Custom Objects

5.1.2.1. Klarna Express Button

The KEB button can be configured and adapted to the needs of your storefront. Steps to enable KEB button:

- To enable the button, your Klarna Merchant ID (MID) and domains must be white-listed. Please reach out to your Klarna delivery manager for support with this activity prior to testing and production go-live (as the domain names white-listed defer). Please ensure that the MID matches that which is configured in the BM service credentials for your storefront.
- To configure the KEB settings for a given site, you must visit “**Merchant Tools – Custom Object Editor**” and search for **KlarnaCountries** custom object.
- Choose the respective country key (e.g.: “US”) and provide the below details:
 - Express Button Merchant ID: *Allow-listed Merchant ID*
 - Express Button environment: *Klarna environment – Production/Playground*
 - Express Button Cart Enabled: *Check box to display on cart page*
 - MiniCart Express Button Enabled: *Check box to display on minicart page*
 - MiniCart Express Button Theme/shape: *theme/shape to display on minicart page*
 - Express Button Library URL: [Available here](#)
 - Express Button Category: *The Klarna category to be pre-selected on Checkout, e.g., “pay_over_time”*
- Please review Figure 45 for reference.

More information to customize the button can be found [here](#).

Note: Klarna Express Button is currently available in [these](#) markets with more to follow

The screenshot displays the Klarna Express Button configuration interface. At the top, there's a navigation bar with 'Sandbox - zzjh', 'RefArch', and tabs for 'Merchant Tools', 'Administration', 'Storefront', and 'Toolkit'. The main configuration area is divided into several sections:

- PDP Placement Data Key:** credit-promotion-badge
- Header Placement Tag Enabled:** ☒
- Header Placement Data Key:** top-strip-promotion-standard
- Footer Placement Tag Enabled:** ☒
- Footer Placement Data Key:** footer-promotion-auto-size
- Info Page Placement Tag Enabled:** ☒
- Info Page Placement Data Key:** info-page
- Data Inline Enabled on PDP/Cart placement (Canada only):** ☒
- URL to On-Site Messaging Library URL:** https://na-library.playground.klarnaservices.com/lib.js

The **Express Button** section includes:

- Express Button Merchant ID:** PNXXXXX
- Express Button Environment:** playground (Playground)
- Express Button Theme:** light (Light)
- Express Button Shape:** rect (Rectangle)
- Express Button Cart Enabled:** ☒
- Express Button Library URL:** https://x.klarnacd.net/express-button/v1/lib.js
- Express Button Category:** pay_over_time

The **MiniCart Express Button** section includes:

- MiniCart Express Button Enabled:** ☒
- MiniCart Express Button Theme:** default (Default)
- MiniCart Express Button Shape:** pill (Pill)

At the bottom left, there is a '<< Back to List' button.

Figure 59 Klarna Express Button completed configuration

Shoppers experience on the storefront cart is as below:

1. Lands on cart page where KEB is displayed
2. Clicks on KEB button
3. Shopper prompted to share their Klarna registered credentials to complete authentication
4. Re-direct to checkout with Klarna Payment Method pre-selected

5.1.2.2. KlarnaCountries

The respective object is dynamically selected based on the request locale country, e.g., SFCC site with locale “**de_DE**” or “**en_DE**” will use the “**DE**” custom object. In cases when the request locale country can’t be dynamically resolved (i.e. with “default” SFCC locale) – attribute “**klarnaLocale**” can be utilized to pass the proper locale to Klarna. For all other cases, this field can be left blank and will not be taken into consideration.

Even if you have locales that are not supported by Klarna Payments, we recommend creating a corresponding entry in the custom object for that locale. Thus, on the billing page of the unsupported locale you will have the Klarna Payments widget showing an appropriate message.

The custom objects store data such as Klarna default locale, service credential IDs and Klarna Payments placement data keys to ensure that Klarna Payments integration is correctly configured.

Note: The same custom object is used by Klarna Checkout cartridge integration!

Administration > Site Development > Custom Object Types > KlarnaCountries - Attribute Definitions

General **Attribute Definitions** Attribute Grouping

Object Type 'KlarnaCountries'

This page lists the attribute definitions of your object type. Use the search to find attribute definitions by ID and name.
Click New to create new attribute definitions. Click Delete to delete existing attribute definitions.

Search Attribute Definitions						
ID or Name	ID	Name	Type	Attribute Settings	Values	
Select All	UUID	UUID	String	*	0	Edit
	country	Country Code	String	*	0	Edit
	creationDate	Creation Date	Date+Time	*	0	Edit
	credentialID	Service Credential ID	String		0	Edit
	klarnaLocale	Klarna Locale	String		0	Edit
	lastModified	Last Modified	Date+Time	*	0	Edit
	psmCartEnabled	Cart Placement Tag Enabled	Boolean		0	Edit
	psmCartTagId	Cart Placement Tag ID	String		0	Edit
	psmLibraryUrl	Library URL	String		0	Edit
	psmPDPEnabled	PDP Placement Tag Enabled	Boolean		0	Edit
	psmPDPTagId	PDP Placement Tag ID	String		0	Edit
	psmUCI	On-site messaging UCI	String		0	Edit

Figure 60 KlarnaCountries Attributes

The table below describes attributes of the **KlarnaCountries** custom object:

Attribute Name	Attribute ID	Description
Country Code	country	Two-letter country code
On-site Messaging Data Default Locale	klarnaLocale	Fallback, if the request locale can't be dynamically resolved, i.e., when using "default" SFCC locale
Service Credential ID	credentialID	The ID of service credentials for this locale.
On-site messaging Data Client ID	osmUCI	The Klarna On-site Messaging "data-client-id" applicable for a given country
Cart Placement Tag Enabled	osmCartEnabled	To enable Cart Placement for a given locale.
Cart Placement Tag ID	osmCartTagId	The Klarna On-site Messaging "data-key" of placement applicable for Cart Page for a given locale.
Cart Placement Theme	osmCartTheme	The theme for cart placement tag. It could be default or dark.
Cart Placement Custom Styling (#osm-cart-placement)	osmCartCustomStyling	Custom styling for cart placement. The css should be wrapped in <style>...customCss...</style> element.
PDP Placement Tag Enabled	osmPDPEnabled	To enable PDP Placement for a given locale.
PDP Placement Tag ID	osmPDPTagId	The Klarna On-site Messaging "data-key" of placement applicable for Product Display page for a given locale.
PDP Placement Theme	osmPDPTHEME	The theme for pdp placement tag. It could be default or dark.
PDP Placement Custom Styling (#osm-pdp-placement)	osmPDPCustomStyling	Custom styling for pdp placement. The css should be wrapped in <style>...customCss...</style> element.

Attribute Name	Attribute ID	Description
Header Placement Tag Enabled	osmHeaderEnabled	To enable Klarna Header Placement in a given storefront
Header Placement Data Key	osmHeaderTagId	The Klarna On-site Messaging “data-key” of placement applicable for the Header
Header Placement Theme	osmHeaderTheme	The theme for header placement tag. It could be default or dark.
Header Placement Custom Styling (#osm-header-placement)	osmHeaderCustomStyling	Custom styling for header placement. The css should be wrapped in <style>...customCss...</style> element.
Footer Placement Tag Enabled	osmFooterEnabled	To enable Klarna footer Placement in a given storefront
Footer Placement Data Key	osmFooterTagId	The Klarna On-site Messaging “data-key” of placement applicable for the footer
Footer Placement Theme	osmFooterTheme	The theme for footer placement tag. It could be default or dark.
Footer Placement Custom Styling (#osm-footer-placement)	osmFooterCustomStyling	Custom styling for footer placement. The css should be wrapped in <style>...customCss...</style> element.
Info Page Placement Tag Enabled	osmInfoPageEnabled	To enable Klarna Info Page Placement in a given storefront
Info Page Placement Data Key	osmInfoPageTagId	The Klarna On-site Messaging “data-key” of placement applicable for the Info Page
Info Page Placement Theme	osmInfoPageTheme	The theme for info page placement tag. It could be default or dark.

Attribute Name	Attribute ID	Description
Info Page Placement Custom Styling (#osm-info-page-placement)	osmInfoPageCustomStyling	Custom styling for info page placement. The css should be wrapped in <style>...customCss...</style> element.
Data Inline Enabled on PDP/Cart placement (Canada only)	osmDataInlineEnabled	Enable this when using PayBright payment method in Canada
URL to On-Site Messaging Library URL	osmLibraryUrl	URL for On-Site Messaging library, applicable for testing or production must be saved. Please use only Klarna production URL in live storefront. Verify test environment URL which includes “playground” in URL For production or live environment, ensure URL includes “production”. E.g: Old library Test URL> https://na-library.playground.klarnaservices.com/lib.js , e.g: Live URL>https://na-library.production.klarnaservices.com/lib.js New library - https://js.klarna.com/web-sdk/v1/klarna.js
On-Site Messaging Environment	osmEnvironment	Environment on which the OSM is used. It could be playground or production.
Express Button Environment	kebEnvironment	The express button environment. Default is playground
Express Button Merchant ID	kebMerchantID	The merchant ID used to display the button

Attribute Name	Attribute ID	Description
Express Button Library URL	kebLibraryUrl	URL of the express button library
Express Button Cart Enabled	kebCartEnabled	To enable the Klarna Express Button on cart page
Express Button Theme	kebTheme	The theme of the button. Options include default, light & dark
MiniCart Express Button Enabled	kebMCEnabled	To enable the Klarna Express Button on minicart
MiniCart Express Button Theme	kebMCTheme	The theme of the button. Options include default, light & dark
Express Button Category	kebCategory	The Klarna category to be pre-selected on Checkout Page, e.g., "pay_over_time"
Klarna Express Checkout Client Key	expressCheckoutClientKey	The Klarna Express client key used for express checkout

Table 7 KlarnaCountries Attributes

The data-client-id and data-key values used in the OSM placements are available in the Klarna Merchant Portal (Europe/US (CA included)/Oceania) within the On-site Messaging App. When selecting the data-key values, ensure that the filter is set to the right country and language.

5.1.2.3. *KlarnaSigningKey*

The respective object is populated when we subscribe to webhooks. For webhook subscription, a signing key gets generated, and this key will get saved to this custom object. The signing key will be later used in code for validating the webhook notification that is received.

The table below describes attributes of the **KlarnaSigningKey** custom object:

Attribute Name	Attribute ID	Description
Signing Key Id	signingKeyId	The signing key id which will be used for subscribing to webhooks.
Signing Key	signingKey	The signing key which will be used for HMAC validation of webhook notification received.

Attribute Name	Attribute ID	Description
Created Date	createdDate	The date when the signing key was created.

Please note that the value in this custom object gets created when ‘CreateWebhook’ job is run. The values in this custom object should not be altered manually.

Custom Objects > Custom Objects > krn:partner:global:notification:signing-key:7a39f909-cd45-4420-943e-76e2d119afc1 - General

General

Manage 'krn:partner:global:notification:signing-key:7a39f909-cd45-4420-943e-76e2d119afc1' (KlarnaSigningKey)

Fields with a red asterisk (*) are mandatory. You can view and edit the name and description in other languages, if required. Click **Apply** to save the details.

Signing Key	
 Signing Key Id: *	krn:partner:global:notification:signing-key:7a39f909-cd45
Signing Key:	FLBwJCT0V1I1+N9pVXXoJPwAk/yN61ms9DLiO+O1fEggk
Created Date:	2025-10-06T06:40:17.748Z

Figure 61 KlarnaSigningKey sample record

5.1.2.4. KlarnaWebhookNotification

The respective object is populated when webhook notifications are received. The notification status will be ‘PROCESS’ initially and once its processed, its state gets changed accordingly. The notification log will contain the notification request and this will be used later while processing the notification.

The table below describes attributes of the **KlarnaWebhookNotification** custom object:

Attribute Name	Attribute ID	Description
Payment Request ID	paymentRequestId	The payment request id will be used to map to the customer who has initiated the payment request.
Notification Log	notificationLog	The notification log will contain the notification request, and this will be used later while processing the notification.

Attribute Name	Attribute ID	Description
Notification Status	notificationStatus	The notification status will be 'PROCESS' initially and once its processed, its state gets changed accordingly.

Please note that the value in this custom object gets created when notifications are received for the payments initiated in the site. The values in this custom object should not be altered manually.

[Custom Objects](#) > [Custom Objects](#) > [krn:payment:us1:request:dbb2fedd-4a82-6e16-b131-3e911e3215f1](#) - General

General

Manage 'krn:payment:us1:request:dbb2fedd-4a82-6e16-b131-3e911e3215f1' (KlarnaWebhookNotification)

Fields with a red asterisk (*) are mandatory. You can view and edit the name and description in other languages, if required. Click **Apply** to save the details.

Klarna Webhook Notification	
 Payment Request ID: *	krn:payment:us1:request:dbb2fedd-4a82-6e16-b131-3e9
Notification Log:	

Attribute Name	Attribute ID	Description
Expiry Time	expiryTime	Timestamp when this data expires (24 hours from creation)

Please note that the value in this custom object gets created only if klarna payments are integratd via PSP. The data gets saved when KEC is initiated from PDP or mini cart or cart pages and also when customer navigates to checkout page.

Custom Objects > Custom Objects > f0b6e9b997a2ff4be802379eba - General

General

Manage 'f0b6e9b997a2ff4be802379eba' (KlarnaInteroperabilityData)

Fields with a red asterisk (*) are mandatory. You can view and edit the name and description in other languages, if required. Click **Apply** to save the details.

Klarna Interoperability Data

Basket ID: f0b6e9b997a2ff4be802379eba

Interoperability Data: [{"content-type":"vnd.klarna.interoperability-data.v3+json","content":{"operation":"payment_request","request":{"amount":"6998","supplementary_purchase_data":{"line_items":{"name":"Men's Resolve Gloves","quantity":1,"total_amount":"6399","total_tax_amount":"305","unit_price":"6399","product_url":"https://zzhj-002.dx.commercecloud.salesforce.com/s/RefArchGlobal/mens-resolve-gloves/061492216683M.html?lang=en_GB","image_url":"https://zzhj-002.dx.commercecloud.salesforce.com/on/demandware.static/-/Sites-apparel-m-catalog/default/dw2f1dad48/images/small/TG733_243.jpg","product_identifier":"061492216683M","line_item_reference":"061492216683M","shipping_reference":"GBP001"}}}}}]

Expiry Time: 02/06/2026 11:54 am
MM/dd/yyyy h:mm a

Figure 63 KlarnaInteroperabilityData sample record

5.1.3. Session Attributes & Cookies

The following session custom attributes are saved in “**session.privacy**” storage and accessible in checkout. The attributes are retained for the session lifetime & cleared when the customer logs out of their profile.

Attribute	Description
KlarnaLocale	The Klarna locale in use
KlarnaPaymentsSessionID (<i>Not applicable since version 21.2.0</i>)	The Klarna session ID returned after “Create Session” API endpoint is called

Attribute	Description
KlarnaPaymentsClientToken <i>(Not included as session attributes since version 21.2.0)</i>	Client token returned by “Create Session” API endpoint and used to initialize the JS SDK
KlarnaPaymentMethods	The available payment method categories for the respective Klarna session; Saved in JSON format
KlarnaPaymentsAuthorizationToken	The authorization token returned by JS SDK “Authorize” call
KPAuthInfo	Whether finalization is required for the payment method; Returned by JS SDK “Authorize” call; Saved in JSON format
KlarnaExpressCategory	The KEB payment category; Currently applicable for US and defaults to “pay_over_time”
KlarnaPaymentsRedirectURL	The URL to redirect the customer to after placing the order; Returned by the “Create Order” API call

Table 8 Klarna Session Attributes

The following cookies are being set by Klarna integration:

Cookie Name	Description
selectedKlarnaPaymentCategory	The selected payment method on checkout (e.g. “pay_later”)

Table 9 Klarna Cookies

5.1.4. Library

In addition to the configurations, the following 2 library assets will be added:

- **“footer-about”** – Updated OOTB asset including link to Klarna OSM dedicated page in the footer.
- **“klarna-email-info”** – Asset containing links to review the Klarna Payment information. Used in the confirmation email sent to the customers.

5.1.5. Services

An HTTP service “*klarna.http.defaultendpoint*” has been added with “*klarna.http.service*” profile and service credentials for each country (described in **KlarnaCountries** custom object).

Please, note that up until version **21.2.0** of the cartridge, the **KlarnaCountries** custom object was **replicable**. To avoid issues with service credentials during replication, merchants should use the same service credential name in staging, development, and production environments!

Please, review section **9.4** Update KlarnaCountries Definition on possible ways to update the definition in your instances.

An HTTP service ‘*klarna.http.webhook*’ has been added for subscribing to webhooks. This service will be used when ‘CreateWebhook’ job is run.

5.1.6. Jobs

5.1.6.1. CreateWebhook

This job is used for subscribing to webhooks. The payment types to which we need to subscribe should be entered in the event types section before executing the job. When the job is run,

- It checks whether a signing key is already present for the storefront site for which it is run. If it is not present, then the signing key is created and saved in the custom object -KlarnaSigningKey. If present, then the signing key id from the custom object will be used.
- Using the signing key id and then event types configured, webhook API is called for subscribing to webhooks.

Once webhook is subscribed, for the subscribed payment types we will start receiving webhook notification which gets saved to the custom object KlarnaWebhookNotification.

5.1.6.2. DeleteWebhookNotification

This job is used for deleting the webhook notifications which are no longer needed. A custom parameter, expiry days, is present which needs to be configured. Any notification which is older than the expiry days configured gets deleted by this job.

5.1.6.3. CleanupKlarnaInteroperabilityData

This job is used for deleting the interoperability data which are no longer needed, and which are expired from the KlarnaInteroperabilityData custom object. The custom object has an expiry time attribute which gets populated based on the logic of 24hrs from the custom object creation time. So, when this job is run, it checks for any entry which is passed its expiry time and then deletes it.

5.2. Logs

The integration includes the following logs:

- Service communication logs – starts with “service-klarna-***”. These logs contain every request and response to the Klarna endpoints. Personal information, i.e. emails & names required for the Klarna API calls are masked in the logs.
- Custom errors and debug info are logged under “customerror-***”, “custodebug-***” & “custominfo-***” files depending on the case.

5.3. Availability

Cartridge functionality will be dependent on the availability of the Klarna API service. Current Klarna operational status can be viewed here - <http://status.klarna.com/>

5.4. Failover/Recovery Process

If Klarna API is not available, Klarna is not presented as a payment option. In case of any failure within the Klarna API, contact Klarna for support.

5.5. Support

A customer service workshop can be conducted during the implementation process before going live to align the operational processes and ensure customer satisfaction. Klarna provides all customers with the possibility to log into Klarna App via website: <https://app.klarna.com/login> or download the Klarna App (free) on a mobile (Android/iOS). The customers can contact support, view their statements, pay for their purchase, track delivery updates, and prolong the due dates if they have chosen to pay after delivery.

5.5.1. Merchant Support

Reporting core SFCC functionality issues in the Klarna cartridge technical integration – please contact commercecloud@klarna.com

For production issue related to Klarna API availability, merchant representative should reach their Klarna Account manager after reviewing the current operational status at <http://status.klarna.com/>. Report the problem in Production (Post Go-live) if you have a suspicion about degraded performances or issues with Klarna's service. The Klarna contact would then be able to report this internally to the incident management team who have established routines to handle and resolve reported incidents. The Klarna contact may request additional information from the individual reporting the problem to help internal team ascertain and identify the issue. The KAM may also advise the merchant to follow the updates on the status page if it is a known incident with on-going updates.

Pre-requisite information to be provided by merchant when reporting incident to help with speedy investigation and resolution:

- Merchant's affected MID or market
- Impact and examples of customer orders (order_id or Klarna session_id if available)
- Screenshots, timeframe, additional information as required

6. User Guide

6.1. Cartridge upgrade

Regular updates to our cartridge code include bug fixes, performance enhancements, security patches, and new features. It's essential to stay up to date with these changes to ensure compatibility, security, and optimal functionality of your integration.

6.2. Upgrade Process

Follow these steps to upgrade your custom code with the latest version of our cartridge:

6.2.1. Review Release Notes

Start by reviewing the release notes for the latest version of the cartridge. The release notes outline the changes, improvements, and any potential compatibility considerations associated with the upgrade.

6.2.2. Assess Custom Code Changes

Identify any customizations or modifications you have made to the SFCC cartridge code in your custom files. These may include storefront customizations, controller adjustments, or custom business logic built on top of our cartridge.

6.2.3. *Backup Custom Files*

Before proceeding with the upgrade, ensure that you have a backup of your custom files, including any modifications made to the SFCC cartridge code. This backup will serve as a safety net in case any issues arise during the upgrade process.

6.2.4. *Compare Code Differences*

Use a version control system or a file comparison tool to compare the differences between your custom code and the latest version of the SFCC cartridge code. Pay close attention to areas where changes have been made to ensure compatibility and maintain functionality.

6.2.5. *Update Integration Code*

Integrate the latest version of the SFCC cartridge code into your custom files, replacing any outdated or deprecated code with the new implementations. Follow the migration guides and best practices provided to ensure a smooth transition.

6.2.6. Test and Validate

After updating your custom code, thoroughly test the integration to ensure that all functionality works as expected. Test various scenarios, including user interactions, data processing, and third-party integrations, to identify any potential issues or regressions.

6.2.7. Address Compatibility Issues

If you encounter any compatibility issues or conflicts with your existing custom code, troubleshoot and resolve them accordingly. Consult our support resources or reach out to our team for assistance in addressing compatibility concerns.

6.2.8. Deploy Changes

Once you are satisfied with the upgrade and have validated its functionality, deploy the changes to your production environment. Monitor the integration closely following deployment to ensure ongoing stability and performance.

6.3. Roles, Responsibilities

There are no recurring tasks required by the merchant. Once configurations are set up, the functionality runs on demand.

6.4. Storefront Functionality

When Klarna has been setup, the Klarna Payments options and iframe widgets will be shown on the billing step. All the SFCC OOTB checkout functionality remains in place, such as but not limited to cart updates during checkout, checkout with applied coupon(s) code(s), checkout with applied product level promotion, checkout with applied order level promotion, checkout with applied shipping level promotion, checkout with applied order level promotion with bonus product.

Select one of Klarna's payment options as the payment method on billing step of checkout process and click the “Next: Place Order” button:

CREDIT

Klarna. Buy now, pay later

Klarna. Buy now, pay later

TESTDRIVE

Buy now, pay later

Pay over time.

3 interest-free instalments

£51.19 pm

Autopay with your card

24 months

£6.47 pm

1st payment

£51.19

15 days later

£51.19

30 days later

£51.21

Learn more

Card Number

VISA 4111 1111 1111 1111

MM/YY

12 / 23

CVC

123

You must be 18+ to use this credit offer. Klarna uses soft searches with credit reference agencies. These do not affect your credit rating. Missed payments can affect your ability to use Klarna.

By continuing I accept the [Pay in 3 terms](#) and [Klarna Shopping Service](#), and confirm that I have read the [Privacy Notice](#) and the [Cookie Notice](#).

Figure 64 Payment Options on Checkout

Depending on the payment method selected and the region, you will see one of Klarna's popup windows to provide the details. Follow the steps on the screen:

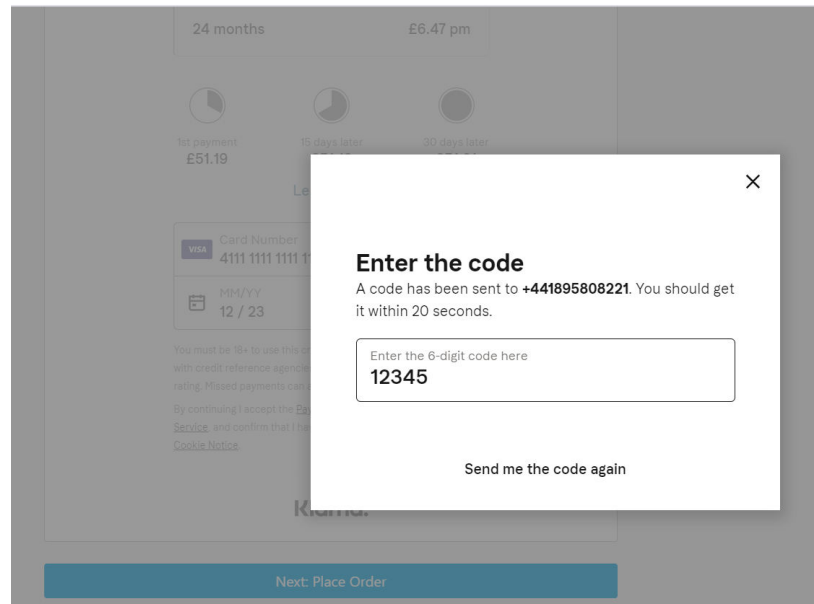


Figure 65 Klarna Popup Screen

On the Review step click on “Place Order” button:

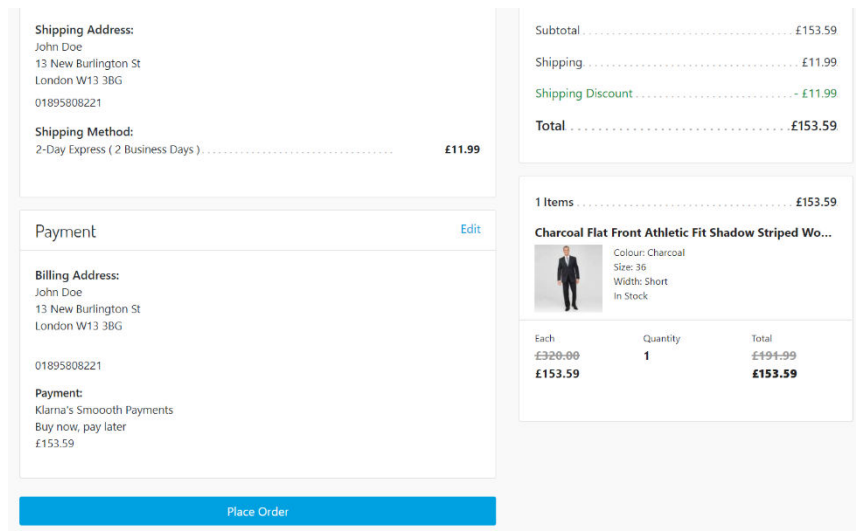
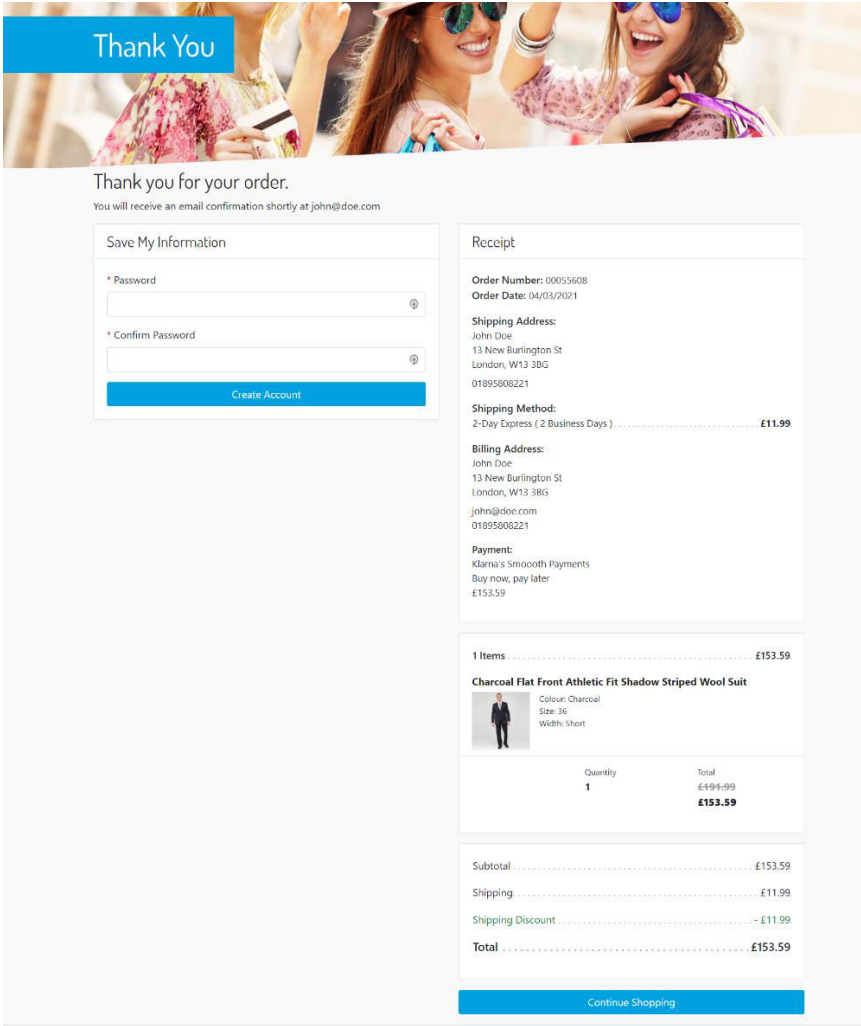


Figure 66 Payment Review Screen

The customer's browser is sent to the “**redirect_url**” and immediately thereafter shown the Commerce Cloud Order Confirmation page.



The screenshot shows a 'Thank You' order confirmation page. At the top, a blue banner with the text 'Thank You' is overlaid on a background image of three smiling women. Below the banner, the text 'Thank you for your order.' is followed by 'You will receive an email confirmation shortly at john@doe.com'. The page is divided into three main sections: 'Save My Information', 'Receipt', and a summary table. The 'Save My Information' section contains fields for password and confirm password, with a 'Create Account' button. The 'Receipt' section displays order details: Order Number: 00055608, Order Date: 04/03/2021, Shipping Address: John Doe, 13 New Burlington St, London, W13 3BG, 01895008221, Shipping Method: 2-Day Express (2 Business Days) for £11.99, Billing Address: John Doe, 13 New Burlington St, London, W13 3BG, john@doe.com, 01895008221, and Payment: Klarna's Smooth Payments, Buy now, pay later, £153.59. The summary table lists 1 item: Charcoal Flat Front Athletic Fit Shadow Striped Wool Suit, with a quantity of 1 and a total of £153.59. The final total is £153.59, and a 'Continue Shopping' button is at the bottom.

Thank You

Thank you for your order.
You will receive an email confirmation shortly at john@doe.com

Save My Information

* Password

* Confirm Password

Create Account

Receipt

Order Number: 00055608
Order Date: 04/03/2021

Shipping Address:
John Doe
13 New Burlington St
London, W13 3BG
01895008221

Shipping Method:
2-Day Express (2 Business Days) **£11.99**

Billing Address:
John Doe
13 New Burlington St
London, W13 3BG
john@doe.com
01895008221

Payment:
Klarna's Smooth Payments
Buy now, pay later
£153.59

1 Items **£153.59**

Charcoal Flat Front Athletic Fit Shadow Striped Wool Suit
Colour: Charcoal
Size: 36
Width: Short

Quantity	Total
1	£494.99
	£153.59

Subtotal £153.59
Shipping £11.99
Shipping Discount - £11.99
Total **£153.59**

Continue Shopping

Figure 67 Order Confirmation Page

The newly created order can be inspected in Business Manager:

Orders

You're using the new Search service.

This page allows you to search for orders by order number. Select **Advanced** to use more search options. Select **By Number** to search by or newline. Entered text is treated as case-sensitive; substring matching isn't supported.

Order Search				
Order Number:			Find	
Number	Order Date	Site	Created By	Registration Status
00055608	3/4/21 4:38:34 pm Etc/UTC	RefArchGlobal	Customer	Unregistered
00055308	3/3/21 6:13:04 pm Etc/UTC	RefArchGlobal	Customer	Registered
00055409	3/2/21 10:07:53 am Etc/UTC	RefArchGlobal	Customer	Unregistered
00055311	3/1/21 10:07:01 pm Etc/UTC	RefArchGlobal	Customer	Registered
00055309	3/1/21 9:55:21 pm Etc/UTC	RefArchGlobal	Customer	Registered
00055307	3/1/21 9:47:04 pm Etc/UTC	RefArchGlobal	Customer	Registered
00055306	3/1/21 6:57:04 pm Etc/UTC	RefArchGlobal	Customer	Registered
00054903	2/19/21 10:10:07 am Etc/UTC	RefArchGlobal	Customer	Registered
00054802	2/18/21 9:15:16 am Etc/UTC	RefArchGlobal	Customer	Registered
00054605	2/10/21 12:33:17 pm Etc/UTC	RefArchGlobal	Customer	Registered

Showing 1 - 10 of 393 items

Show [50](#) [100](#) [All](#) items

Figure 68 Orders List in BM

Klarna Payments order id can be inspected in the Attributes tab of the order:

[Merchant Tools](#) > [Ordering](#) > [Orders](#) > Order: 00055608(RefArchGlobal)

[General](#) [Attributes](#) [Payment](#) [Notes](#) [History](#)

Attributes for Order '00055608'

On this page you can edit the attributes of the order. Fields with a red asterisk (*) are mandatory. Click **Apply** to save changes. Click **Reset** to revert your changes.

Klarna Payments	
Klarna Payments Order ID:	6fa5c3ad-411d-2629-b543-4f5ea672ea9e
Is VCN Used:	☐
VCN Card ID:	

[<< Back to List](#)

Figure 69 Order Attributes

Payment method details can be inspected on the Payment tab of the order, and it should be Klarna:

[Merchant Tools](#) > [Ordering](#) > [Orders](#) > Order: 00055608(RefArchGlobal)

General	Attributes	Payment	Notes	History
---------	------------	----------------	-------	---------

Payment Information for Order '00055608'

Order Total:	£153.59
Amount Paid:	£0.00
Balance Due:	£153.59

Invoice Number:	00242008
Payment Status:	Paid

Payment Method:	KLARNA_PAYMENTS Processor: KLARNA_PAYMENTS Transaction: 6fa5c3ad-411d-2629-b543-4f5ea672ea9e Amount: £153.59 Klarna Payment Category ID: pay_over_time Klarna Payment Category Name: Buy now, pay later Fraud Status: ACCEPTED
-----------------	--

[<< Back to List](#)

Figure 70 Order Payment Detail

Order can be further inspected in Klarna Merchant Portal:

- EU: eu.portal.klarna.com
- US: us.portal.klarna.com
- OC: us.portal.klarna.com

#7BRVXW4H

Captured

£153.59

Merchant reference 1

00055608

Edit

Merchant reference 2

cf8b55f8f98bc392030178105

Edit

Created

Mar 4, 2021, 6:38 PM

Expires

Apr 1, 2021, 3:00 AM

Merchant ID

K500726

Customer

Shipping address

John Doe

13 New Burlington St

London

W13 3BG

GB

Tel

01895808221

Email

john@doe.com

Edit shipping address

Billing address

Additional Info

Order lines (2)

Refund

Print packing slip

	Item / Reference	Qty	Unit price	Discount	Tax	Amount
☐	Charcoal Flat Front Athletic Fit Shadow Striped Wool Suit 640188017003M	1	191.99	38.40	5% 7.31	153.59 Captured
☐	2-Day Express GBP002	1	11.99	11.99	5% 0.00	0.00 Captured

PAYMENT DETAILS

Initial Payment Method

Pay later in parts

VISA 411111*****1111

Resend statement

ORDER TOTAL

£153.59

Captured

£153.59

Refunded

£0.00

Not Captured

£0.00

CUSTOMER BILLED

£153.59

Activity Log

Mar 4, 2021

8 minutes ago

6:38 PM

Captured: £153.59

Via API

6:38 PM

Order placed: £153.59

By Klarna

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Terms & Conditions

Figure 71 Klarna Portal Order View

7. Known Issues

The LINK Cartridge has no known issues.

8. Release History

Version	Date	Changes
18.1.0		Initial release of Klarna Payments SFRA.
19.1.0		Added SFRA version
19.1.1		Updated VCN to use the newest API version
19.1.2		Fix auto capture for the pipelines cartridge
19.1.4		New country locales added. Minor bug fixes. Cartridge templates and forms updated for latest SFRA.
19.1.5		Added additional verification for all notifications. Minor fixes around the configuration objects. Added Canadian support. Documentation updates.
19.1.6		New country locales added. Updated VCN to store encrypted card details
21.1.0		Fixes around discounts taxation & VCN error handling. Added VCN improvements, additional OSM placements, BOPIS support. New IT, CA, FR & NZ country locales. Removed acknowledge call. Documentation updates.
21.1.1		New On-Site Messaging attribute for Canada. Remove not required locale templates for SG Spain & Belgium. Documentation updates.

Version	Date	Changes
21.1.2		Fixed core file naming convention issues in 21.1.0 and 21.1.1. Please upgrade to the latest version if you are currently using 21.1.0 or 21.1.1. Removed deprecated “scripts/util/Builder.js” file.
21.2.0		Added Klarna Express Button. Moved Klarna session ID & client token from SFCC session privacy to Basket attributes. Changed KlarnaCountries definition to not replicable. Code cleanup. Documentation updates.
21.3.0		Improvements for create_session errors Expired user session issues related to empty shipment.shippingMethod Additional locale (PL) included in config files
21.3.1		Documentation updates.
22.1.0		Improvements for create and update session errors Added Klarna Express Button in minicart. Support for long running basket Rate-limits by operations
22.2.0		One Klarna Optimisation Mexico locale support
22.2.1		Rollback of One Klarna Optimisation
22.3.0		SFRA ver. 6.0.0 support Rollback hide VAT from Checkout functionality

Version	Date	Changes
22.3.1		Fix User-agent version sent to Klarna services
22.4.0		Intent field addition in Klarna Payment session creation Combine Klarna Authorization and Create Order in Checkout Review Step
22.5.0		OMS support
23.1.0		Fix issue with incorrect values for EMD Improvement Klarna Auto Capture and error handling Logging information for troubleshooting bugs Add Auto_finalise=True to the review checkout flow
23.1.1		Fixed an issue where sessions with negative order_tax_amount occurred due to SFCC session expiration Compatibility mode 21.2 support Replace deprecated window.KlarnaOnsiteService.push with window.Klarna.OnsiteMessaging.refresh.
23.2.0		Subscription Payments support: recurring payments and subscription handling directly within the SFCC environment. This update includes configuration options, subscription management in the cart and checkout pages, and a customer dashboard for subscription oversight. Klarna Bank Transfer payments: added a new server-side authorization callback feature for Klarna Bank Transfer payments, enhancing reliability across EU markets and supporting all existing KP cartridge functionalities.
24.1.0		Klarna Express Checkout: a new feature introduced in Storefront where Express Checkout Button will be displayed in PDP, Cart and Mini Cart and user will be redirected to Klarna after clicking Express Checkout button. Klarna Express Checkout enabled quick and easy checkout where Shipping Address, Billing Address and Payment details will

Version	Date	Changes
		be preselected so that Checkout can be completed in fewer clicks. Fix for create order service calls with 500 status response: now orders are not created and error is thrown for these cases.
24.2.0		Update of OSM functionality that is more aligned with current web standards, provides a consistent identifier system as used in KEC, and offers enhanced customization options to our merchants. This will not only improve the user experience but also reinforce Klarna's commitment to providing versatile and state-of-the-art e-commerce solutions. CSS customizations are available only in the new library version. Documentation update - new section added for cartridge upgrade process.
24.3.0		Update of Klarna Express Checkout configuration. The display of Klarna Express Checkout buttons is now customizable by selecting preferred placements. By default, none is selected. Documentation update – new section added “Klarna Display Conditions and Authorization Handling”.
25.4.0		Added webhook subscription for KEC one-step checkout via PSP. A storefront site can now subscribe to webhook and receive notifications for subscribed payment events. These notifications are stored in a custom object for further processing. Documentation update – Updated new custom objects 'KlarnaSigningKey' and 'KlarnaWebhookNotification'. Updated services section in which API related to webhook is added. Added a new section for jobs, CreateWebhook and DeleteWebhookNotificaiton.

Version	Date	Changes
25.5.0		Added interoperability_data to the session. This data includes the order amount and the order lines for which the payment was initiated. The PSP will be able to retrieve this information from the session for further processing.
25.6.0		Upgraded OSM to use Klarna Web SDK version 2. Adding KEC one step for direct merchants. Added metadata in every request to Klarna, be it in Web SDK initialisation or in Api calls.
26.1.0		Updated interoperability data storage to use a custom object instead of session storage. Enhanced OSM installment messaging to dynamically refresh on the checkout page when the purchase amount changes.
26.1.1		Added field validation on the Client ID site preference to ensure only allowed values can be entered.
26.2.0		Added support for passing currency and correctly mapped theme parameters in Klarna OSM API requests. Introduced Sign in with Klarna Web SDK v2, restoring updated SIWK flow and replacing refresh token usage with ID token authentication. Removed legacy Klarna Web SDK v1 script and ensured only v2 SDK is loaded across all pages to improve performance. Updated Klarna integration to replace deprecated interoperability_token with klarna_network_session_token across KEC and standard checkout flows. Fixed Klarna callback failures with SFCC session filtering enabled by replacing queryOrder() with direct order lookup

Version	Date	Changes
		and introducing custom object mapping for reliable server-to-server processing.

9. Additional Information

9.1. Klarna API Information

The Klarna Payments API is accessible through different endpoint based on the context of the webstore. There are separate endpoints for testing and live and the Klarna merchant identifier (MID) is configured for respective markets in regions (EU, NA, OC) by endpoint.

9.1.1. Live Environment

The API for the European production environment can be found at

- <https://api.klarna.com/>

The API for the North America production environment can be found at

- <https://api-na.klarna.com/>

The API for the Oceania production environment can be found at

- <https://api-oc.klarna.com/>

9.1.2. Testing Environment

The API for the European Playground/testing environment can be found at

- <https://api.playground.klarna.com/>

The API for the North America Playground/testing environment can be found at

- <https://api-na.playground.klarna.com/>

The API for the Oceania Playground/testing environment can be found at

- <https://api-oc.playground.klarna.com/>

9.2. Generate Key Pair and Key Id for Virtual Card Settlements (VCN)

The recommend RSA keypair size of 4096 bits. This key pair must be associated with a key_id (UUIDv4). The public key must be shared in JWK format with Klarna contact. Note that for

production and playground, the `key_id` and keypair combination shared are different and must be configured prior to testing/go-live of the virtual card product.

To generate an RSA keypair with a 4096-bit private key you can use the following *openssl* command:

```
openssl genpkey -algorithm RSA -out private_key.pem -pkeyopt rsa_keygen_bits:4096
```

To extract the public key from an RSA keypair, you can use the following *openssl* command:

```
openssl rsa -pubout -in private_key.pem -out public_key.pem
```

In the folder where you have executed the above commands two new files will be created - **public_key.pem** and **private_key.pem**.

The contents of the files should look something like:

public_key.pem

```
-----BEGIN PUBLIC KEY-----
MIICljANBgkqhkiG9w0BAQEFAAOCAg8AMIICCgKCAgEAOYNG7l2G8nZa+22oBYZk
tV228lw3UE9WO4oxfknJtKEdHn84x55ULt8KQTh9NVtdeKC8nTfTgyvMt/GNCa18
xuZV/IGYDftKt85hbV5EjOum+StAIufEXvIBX7nMOMc1KyWm9kp2kbqd88mFIX63
KV94OoNEXcNatRDFYR+qz53+ifadDQtQ1slVNStdroCZDJ1+LxtBy9V+BdmsBK1E
RLsKh/JLXyWE24FJKV+z00s7TQkdWW/5ET12OGQYZsWo1yqgi9HplNvrisve8vWP
xaL4m8iZ3I/9yYdg7yANQbTxSJcbbRCgaaagPo30CNxeqU6qafY5g8vY3E52CoXH
DdO4UslX1qcuYIDhqaDzey6W+b8m755xLi+rqQyM4PBWL0J0dM3FVid8+4YKILex
3AKBFciqRCMHsOGaEeyrXKTjlAsghr9RS8PifvQRrL440cHzqw2vX0DvpjSWcmUJ
tW4wUq5RNSsobrxnVmoV6fj1z67Q/1P+l5Ie+oowdahR5ztVqJlO+2PNoX4I5VDs
/Pkz3f8wWVc3Mp2oNT244o+/NIiyRfPFaJJx7JAgrcvZt2nFamY4QApXLFJCpgEM
wYucE4AH4gJKsh3KZbxRERrrO72bL2rxvWqBp/0h7DcMsV9sQs4BvxxIl6CF506F
ThzmclaKLBAyd5LALiXiPfkCAwEAAQ==
-----END PUBLIC KEY-----
```

private_key.pem

-----BEGIN PRIVATE KEY-----

MIIJQQIBADANBgkqhkiG9w0BAQEFAASCCSswggknAgEAAoICAQCglgbuXYbydlr7
bagFhmS1XbbyXDdQT1Y7ijF+Scm0oR0efzjHnlQu3wpBOH01W114oLydN9ODK8y3
8Y0JrXzG5lX+UZgN+0q3zmFtXkSM66b5K0Ai58Re+UFfucw4xzUrJab2SnaRup3z
yYUhfrcpX3g6g0Rdw1q1EMVhH6rPnf6J9p0NC1DWyVU1K12ugJkMnX4vG0HL1X4F
2awErUREuwqH8ktfJYTbgUkpX7PTSztNCR1Zb/kRPXY4ZBhmxajXKqCL0emU2+uK
y97y9Y/FovibyJncj/3Jh2DvIA1BtPFIlxttEKBppqA+jfQI3F6pTqpp9jmDy9jc
TnYKhccN07hSyVfWpy5ggOGpoPN7Lpb5vybvnnEuL6upDIzg8FYvQnR0zcVWJ3z7
hgogt7HcAoEVyKpElwdl4ZoR7KtcpOOUCyCGv1FLw+J+9BGsvjjRwfOrDa9fQO+m
NJZyZQm1bjBSrlE1KyhuvGdWahXp+PXPrtD/U/6Xkh76ijB1qFHnO1WomU77Y82h
fgjlUOz8+TPd/zBZVzcynag1Pbjij780iLJF88VoknHskCCty9m3acUCZjhAClcs
UkKmAQzBi5wTgAfiAkqyHcplvFERGus7vZsvavG9aoGn/SHsNwyxX2xCzgG/HEiX
oIXnToVOHOZyVoosEDJ3ksAuJeI9+QIDAQABAoICACRkaUsUNI22RB3yEPu3DiCP
pO6v+QAeA4gTW+GUdqR9dCZLaSCZ7bhxVVOuoX4qPzslO6hjUmOyzG6upFgVPk+P
HNQfyEUZoC148Eib9OziAXUN2URMpv1KbwVm+BO814X8zguai7uru0PHTG1oy677
4Ct1OknxAxxHQDIaxT6XJFo5SA4EinUfNz2Bo3/xry/QjxW/mCK0GwDd4PNp9TGM
FPTv2SgdSDOWzGQlOH5N3owuzMpI8NV6z74wv+i5Ptv41Dzu8WhyXpiYsSk00SRK
HPC68j2bAzTPghp5aSZ9976SGm2SPonJXyboXdiHbI/osdyqDxeIT3iB9GmrHX/i
kHPGJCh7fRZvqj39Hc+IxYjabwW3rDeDIPB7ab9z1KLF4z1D6AZOKCPyTaDRdQ1Q
eDi7LwDmk7NHEPrmF/nlcpuQdqblbmFO2zEs0TOe6y4uBMndRsbQprTNSMudBkrA
INaYVSTQ1Z0Y/8DZDpGcyS1OnJv74F15uDjKN6/ov991mZ1JrZ+V2sdS3EDUImvP
6thQKwI7Ln6h+ApHtWUG1NmvQe5gJE0qAeJ9b45clUzIRUwhVmEp8NoIJh0kAjaN
d4lk7xy9ZRDUY5yekPeYrJPShjsHAyEoktJIjRufI2UUq3uxNjjICoQcOVGfNDIS
YTTPwpulpmC0C+rh2fgBAoIBAQRultRARvte2JkhVOUyZk88zd9kvrI6fNiyKmi
HgiWf7qkTPD9xhOQWDw3iwrFQAD+YkgV5MCBO8wp8oO8GEOCI+XZWEExOcPT0
Vfj
PZHiQrTFnlFG/+fAO14xLf3j3ED4YQXdhOKI3xoLknQx/EydLoctxgkkgpWLrsA7
DwdSAgl/0sBvaHY27ogAfdimHdaKZ5OAe4a9k1qP3xVZBuOe8Sd65unBavUJLDuv
ikeNmkSVgW1sm55/729Jlr63USHF76It+vElcdZ+vKg5vYotsQgPzvNBmUO/E8Gj

zMXQRfqvEDINXEX0rCupTkwlG6AGTwQc/NPzyr/LTpLe6UBAoIBAQDEUjTiG11V
hf7WjdG3gctRlr+mYapQHgXdVLx2QSaQUYid+0QXK11YfJlsRB6nwa+OED83RfP0
lIFqxpzudSLPmoDuIBT7Dl5c/aleyKs/siUusP8QVDXk6OAR84XSytC35sIRV7pE
VMuBL91jfkQ0Lf/PreslK/ki6Yvwwp4qrHK6/f9TgciHclYtf+/oti4ky6GJgfmP
fmuCqjxmUKbXXFPd5RbL2THGOowilb8zDLjf3RlbiQFqogAk6H9hp2V0VZLiJHp
UWM3z3zxDWeDaqJ08sHuk/rA9QpsVTu8IGTQsxdj8JwluN1Q+YZiOuPiSENBqPzT
V3exexo3sD5AoIBAGU3qEyPojz1+9D1Sa8LW2CABzlq4z9g84ABAZOslxX5q7W
x1PinZyDSQSRXglB13jt29ZdIR79ygnQlglYOBjcvtgVQHPuafk3RlBQbbCh+val
9dn/tUxMGqnhnKaby1rovJHfdqnPpKwzNAjYUqaGkJ822xhmmke/fEyAanIPa4
stDRvIPEWPTLx5xcOCdx13khpKSngkRvaLEfpwkVX7Vr7hK/2OSFaYTNmrzXYBQ7
c6D/9d3Oo4nLb/mu+Tq67S19t53Qg/GEgTfkpuRoVPi0KyhUnKKCGWIBMZLTwyIG
S9eTFDKoJ0cSTGipjW7bPua93wZ8eEbRABpf4QECggEANNhQBeEJ0aCdBVHtdrEI
crDaa8X0W1aJi5dol4hYCRajaKsfHAF/QfdgMQVxHwUC5YG4En/Q+DAVWhGWYpXD
RhC3zeFy5FVszyk0sx/fAOIKGvRn5BRW4YRR9GMRzbjsT+RcruBnckdE9ERXGpX9
c/JB3rxZBI+oliFM8yfWKtMwsrmNKtFuDftvJeok4KejycFF4eWDqsf828xjPT+
xA/FP4CQD1UqkcpmuFSIwAwXo6LXVY7NTS0nKMiUnTLkLlTIHtLnO9+9jmNapWRP
Tc+hZUuHKlpI8DHFmX2j87LgkFD05eD5lynY4RgZtU1W1C1RdVYwoA72WB7knEaB
uQKCAQAH9s67P/7fFX9dfEans3PHU4nGjD8dJ8eoNQ6DhBMydZpGWI5ZUeEBZDRk
0cBOeRs5BOcS43Em9kETpzawyCwxmnwzl+CzoPzMqCTw9tXomF9HG6RJ9XBdJfGA
ALAwCd4bASxmFM6guSP5GKnZ9aY3tR3tWWDfr7f9z8wOewzzpPclwRh009fPe4TC
NXoEm1MELJVeUieDSLKZgjcW8WHGqLlONpA0/fwSM2gIcxETVV7qx3aPuJzCVh
LQZoBLQk3UMKsWDdpzeBdiERe66NAGVk92Xe7SY9EY2vymaq761ilxlvprT27qp
240LDJawqM0IraKmdCvWjofWSaOU
-----END PRIVATE KEY-----

9.3. Decrypt VCN Card Details

To decrypt the virtual card details stored on order level and authorize the credit card processor you can use the following code snippet. You can find more information about the decryption process [here](#).

```

var OrderMgr = require( 'dw/order/OrderMgr' );
var Cipher = require( 'dw/crypto/Cipher' );
var Encoding = require( 'dw/crypto/Encoding' );
var Site = require( 'dw/system/Site' );

var Order = OrderMgr.getOrder( "order_id" );
var VCNPrivateKey = Site.getCurrent().getCustomPreferenceValue( 'vcnPrivateKey' );
var cipher = new Cipher();

var keyEncryptedBase64 = Order.custom.kpVCNAESKey;
var keyEncryptedBytes = Encoding.fromBase64( keyEncryptedBase64 );
var keyDecrypted = cipher.decryptBytes( keyEncryptedBytes, VCNPrivateKey,
"RSA/ECB/PKCS1PADDING", null, 0 );
var keyDecryptedBase64 = Encoding.toBase64( keyDecrypted );
var cardDataEncryptedBase64 = Order.custom.kpVCNPCIData;
var cardDataEncryptedBytes = Encoding.fromBase64( cardDataEncryptedBase64 );
var cardDecrypted = cipher.decryptBytes( cardDataEncryptedBytes, keyDecryptedBase64,
"AES/CTR/NoPadding", Order.custom.kpVCNIV, 0 );

var cardDecryptedUtf8 = decodeURIComponent( cardDecrypted );
var cardObj = JSON.parse( cardDecryptedUtf8 );
var expiryDateArr = cardObj.expiry_date.split( "/" );

// Retrieve ecnrypted card details
var cardPAN = cardObj.pan, cardCVV = cardObj.cvv,
    cardExpiryMonth = expiryDateArr[0], cardExpiryYear = expiryDateArr[1];

```

9.4. Update KlarnaCountries Definition

With version 21.2.0 of the cartridge, the KlarnaCountries custom object definition changed to non-replicable. If you are using an earlier version, to mitigate any issues that may be present in your environments, please follow these steps to update the definitions:

1. Back-up KlarnaCountries configurations per country for each site as changing the definition will remove all configurations on your environment!
 - a. For merchant with one site - Go to "**Merchant Tools > Custom Objects > Import & Export**", select KlarnaCountries & export the data.
 - b. For merchant with multiple sites using Klarna – You can export the configurations via "**Admin > Site Development > Site Import & Export**" and select each site that uses Klarna to be included in the export zip file. This section requires Account Manager access for users and will export all custom objects, not just Klarna ones.
2. Export custom object definitions from "**Admin > Site Development > Import & Export**". This action will export all custom object definitions that you have.
3. Update KlarnaCountries.xml definitions in file exported in step 2 and set "**<staging-mode>source-to-target</staging-mode>**" to "**<staging-mode>no-staging</staging-mode>**"
4. Import the updated KlarnaCountries.xml definition in "**Admin > Site Development > Import & Export**".
5. Import the KlarnaCountries configs exported in step 1 (manually or via bulk import)
 - a. If you've followed step 1.b with multiple sites, you may want to edit the zip file and remove everything else apart from KlarnaCountries.xml